

ALGORITHMIC TRADING IN FINANCIAL MARKETS: A STUDY OF PROFITABILITY OF THE POPULAR TECHNICAL ANALYSIS ALGORITHMS

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ABSTRACT

In algorithmic trading, computer systems replace human traders by making trading decisions based on mathematical models to identify profitable trading opportunities. This study investigates the profitability of mathematical models used in algorithmic trading systems to perform intraday technical analysis in the London equity markets.

Six different models were employed: Naive, Simple Moving Averages, Exponential Moving Averages, Moving Average Convergence and Divergence (MACD), Relative Strength Index, and Kalman Filters. The models were tested on one minute interval intraday data on five large cap stocks which are constituents of the FTSE100 Index. Transaction costs, out of sample verification, amongst other issues were taken into consideration. A non parametric bootstrap was used to confirm the mean returns were statistically different from zero. The Sharp Ratio was calculated to rank the returns produced by the models given their relative riskiness.

The study found that the MACD performed the best and was able to give statistically significant returns on the data set chosen. The evidence of profitability of the MACD model in the London Market goes against the Efficient Market Hypothesis proposed by Fama and supports the alternative models based on behavioural finance which state that the markets have short term momentum which can be exploited to make statistically significant returns.

CONTENTS

Abstract	i
List of Figures	iv
List of Tables	v
Introduction	1
CHAPTER	PAGE
1 Background: Algorithmic Trading	3
1.1 The Rise of the Machines	3
1.2 Algorithmic Trading Components	4
1.3 Alpha Model: Technical Analysis	6
2 Literature Review	8
2.1 Theoretical Literature	8
2.1.1 The Efficient Market Hypothesis	8
2.1.2 Alternative Models	11
2.2 Empirical Literature	12
2.2.1 Technical Analysis on intraday Equity Markets	12
2.2.2 Evaluation of Technical Analysis	13
3 Data and Empirical Methodology	17
3.1 Testing Mechanism	17
3.2 Data Examination	18
3.2.1 Descriptive Statistics	20
3.3 Technical Trading Rules	21
3.3.1 Naive Filters	22
3.3.2 Simple Moving Average Filters	22
3.3.3 Exponential Moving Average	23
3.3.4 Moving Average Convergence/Divergence	24
3.3.5 Relative Strength Index	24
3.3.6 Kalman Filters	25
4 Empirical Findings	26
4.1 Summary Results	26
4.2 Analysis	28

5	Conclusion	30
APPENDIX		PAGE
A	Appendix	32
A.1	Acknowledgement	32
A.2	Training Data Statistics and Graphs	33
A.3	Summary Results	40
A.4	Test Set Statistics And Graphs	43
	References	49

LIST OF FIGURES

Figure		Page
1.1	Algorithmic Trading System Components (Nutti et al. [2011])	5
1.2	Alpha Model (Treleaven et al. [2013])	6
A.1	BLT (BHP Billiton) Training Set	34
A.2	DGE (Diageo) Training Set	35
A.3	GSK(GlaxoSmithKline) Training Set	36
A.4	HSBA (HSBC Holdings) Training Set	37
A.5	TSCO (Tesco) Training Set	38
A.6	Technical Indicators DGE Intraday Prices (8/6/2015)	39
A.7	BLT (BHP Billiton) Test Set	44
A.8	DGE (Diageo) Test Set	45
A.9	GSK(GlaxoSmithKline) Test Set	46
A.10	HSBA (HSBC Holdings) Test Set	47
A.11	TSCO (Tesco) Test Set	48

LIST OF TABLES

Table		Page
3.1	Randomly Selected Stocks	19
4.1	Summary Results of all Models	26
4.2	Results with Broker Fees 0.9 cents	27
4.3	Summary Results of Models vs Ticker	27
A.1	Training Set Statistics	33
A.2	Summary Result for the EMA Model	40
A.3	Summary Result for the Kalman Model	40
A.4	Summary Result for the MACD Model	40
A.5	Summary Result for the RSI Model	41
A.6	Summary Result for the SIMPLE 1 Model	41
A.7	Summary Result for the SIMPLE 2 Model	41
A.8	Summary Result for the SMA Model	41
A.9	Summary of Performance by Date	42
A.10	Summary Time Ticks	43

INTRODUCTION

Algorithmic trading has seen rapid growth in recent years. In 2011 it accounted for 73% of the U.S. equity volume (Treleaven et al. [2013]). In algorithmic trading, computer systems replace human traders by making trading decisions based on mathematical models to identify profitable trading opportunities.

The mathematical models perform technical analysis in the pre trade component to analyse real time or historical data to identify the profitable opportunities (market timing). The efficient market hypothesis proposed by Fama [1970] claims that technical analysis should not be able to generate profits from market timing.

Practitioners who use technical analysis for trading believe that the price histories contain information about the future behaviour of the financial asset. The methods used in the technical analysis approach are not the only concern for researchers and practitioners. Without sound data, researching efficient algorithms for trading becomes more difficult. The raw data used by these models is hugely expensive and cleaning the data is highly time consuming, but essential due to the sensitivity of the trading algorithms (Treleaven et al. [2013]). Also details of the order book and cost of trading are an important considerations.

The objective of this study is to investigate the intraday profitability of mathematical models used in technical analysis in algorithmic trading systems on London equity markets. The models are tested on a subset of stocks which are constituents of the FTSE100 Index. These stocks are very liquid with high intraday volumes. Technical analysis is performed on the price data with one minute interval available from Google Finance. Six different types of technical analysis algorithms are tested Naive, Simple Moving Averages, Weighted Moving Averages, Moving Average Convergence and Divergence, Relative Strength Index, and Kalman Filters. The study found that the algorithms based on moving averages performed the best intra day on the selected data set outperforming both a buy and hold strategy and naive filters.

The thesis is organized as follows: Chapter 1 details the background of algorithmic trading their use of technical analysis. In Chapter 2 details of theoretical literature on the subject and past empirical studies are presented. The data and methodology of the study is discussed in Chapters 3. Chapter 4 contains the analysis of the results and Chapter 5 concludes the study.

BACKGROUND: ALGORITHMIC TRADING

Technological advancements in the trading process have extensively changed the dynamics of financial markets. Each step in the trading process has become highly automated, dramatically changing the way financial assets are traded. The automation of the trading process with the advent of sophisticated computer technology has increased the complexity. One of the key outcomes of this has been the development of Algorithmic Trading (AT) in the financial markets. This Chapter details the background on how algorithmic trading, importantly describing in detail the components of an algorithmic trading system and its types. The Chapter concludes with a details on technical analysis and its use in algorithmic trading systems which is the topic of this study.

1.1 THE RISE OF THE MACHINES

According to Nuti et al. [2011], “An intuitive way to classify algorithmic trading is through the separate processes being automated within a trade’s life cycle”. In these separate processes intricate analysis and decision making is performed by preprogrammed computer algorithms which act without human intervention. A few key factors have contributed to the rise of these AT machines.

The first was the introduction of Electronic Communications Networks (ECNs) in the 1990s. The ECN allowed trading of financial securities outside of the regular exchanges. Market participants could subscribe to these systems and enter orders electronically. This technological revolution enabled the participants to remotely access multiple execution venues without the need for physical presence in the exchange. Steady investment made into the ECNs improved speed and efficiency which lowered costs and manual errors. However only two exchanges, NYSE and NASDAQ, were the beneficiaries. In 1998, the Securities and Exchange Commission (SEC) issued the Regulation

Alternative Trading Systems (RATS) which resulted in the creation of a number of alternative electronic trading platforms(Agarwal [2012]).

The technological revolution in the financial industry was not the only factor contributing to this growth. In early 2000, the US stock exchanges narrowed the spreads for bid and ask by moving away from fractional quote system to decimal based quotes. This reduced the minimum spread between the bid and the ask from $\frac{1}{6}^{th}$ of a dollar to 1¢. Traders which relied on making a profit through bid/offer spreads based on the old quotes needed better alternatives which gave rise to AT(Agarwal [2012]).

In 2005 another important regulation made an impact on the rise of AT. The SEC in an attempt to promote transparency and competition passed the Regulation National Market System (RNMS) which updated the Trade-through Rule. This ruling allowed trade orders to be posted nationally and not at individual exchanges enabling traders to leverage and profit from price discrepancies between different exchanges(Agarwal [2012]).

As mentioned earlier, AT accounted for over 73% of the U.S. equity volumes in 2011(Treleaven et al. [2013]). A report by the Senior Supervisors Group (April 2015) states that the role of traders have changed extensively since the development of AT, and leading firms that engage in AT have exceedingly benefited from these changes. They argue that traders now use trading strategies based on algorithms which they use to arbitrage price differences and take advantage of liquidity or lack thereof, as well as, alter the level of aggressiveness of trades by tuning these algorithms.

1.2 ALGORITHMIC TRADING COMPONENTS

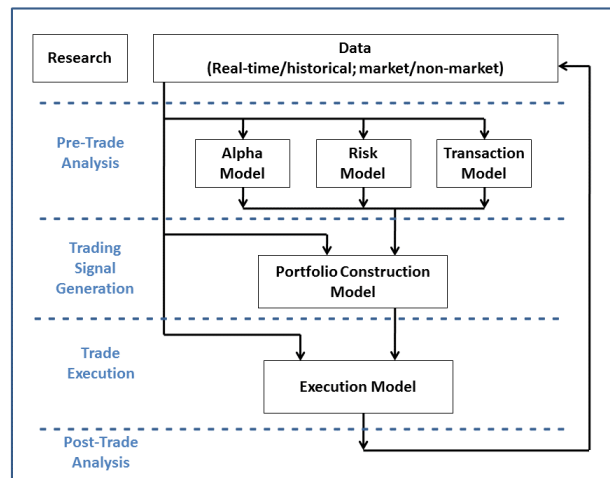
AT can be described as the use of computer algorithms to make trading decisions without human intervention. These computer algorithms are simply a set of step by step instructions or rules that take form of a computer program. These algorithms determine the time when the trade is to take place, at what price it should take place and at what quantity, while simultaneously monitoring market conditions, reducing market impact and closely tracking benchmarks over the execution interval (Hendershott et al. [2011]).

AT is often confused with other terminologies such as electronic trading, order-management system, and automated trading.

- Electronic trading systems allow for the exchange of financial assets such as cash equities, bonds, and foreign exchange electronically between market participants.
- Order management systems enable participants to manage order execution by connecting them to one or more electronic exchanges
- Automated trading systems generally refer to trade execution programs that automatically submit trade orders to an exchange.

The components of the trade life cycle automated by AT system are shown in the Figure 1.1 below. The components are categorized into five main components: data access/cleaning, pre trade analysis, signal generation, execution, and post trade analysis(Nuti et al. [2011]). The data access and cleaning part obtains financial data from feeds and other sources. It systematically cleans the data making it ready for pre-trade analysis. The pre trade analysis components identify trading opportunities using the cleaned market data and the trade signal generation component identifies the portfolio of assets to be accumulated. The trades are then executed by the trade execution components, and the post trade analysis looks at the affects of the trading actions in the market (Treleaven et al. [2013]).

Figure 1.1: Algorithmic Trading System Components (Nuti et al. [2011])



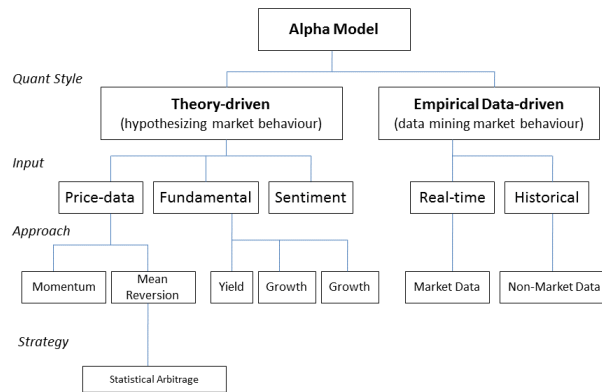
In pre trade analysis component the system analyses properties of assets through data analysis to identify trading opportunities. These properties are identified through three mathematical models:

alpha, risk, and transaction cost. The risk model measures the financial risk level associated with the assets. The transaction cost model calculates the potential trading costs of the assets. While the alpha model is designed to predict the future behaviour of financial assets that the system is designed to trade (Nutti et al. [2011]).

1.3 ALPHA MODEL: TECHNICAL ANALYSIS

The alpha model predicts the future behaviour of assets through analysing real-time and historic data to identify potential trading opportunities. The analysis in Alpha model as shown in Figure 1.2 is divided in to two basic strategy approaches: theory driven or empirical driven. In the theory driven approach a preconceived hypothesis describing the behaviour of the securities is confirmed using modelling. While in the empirical approach, data mining algorithms are used to identify patterns in the data without having views about the behaviour of the underlying security (Treleaven et al. [2013]).

Figure 1.2: Alpha Model (Treleaven et al. [2013])



In the theory driven approach fundamental, technical, and quantitative are the three main categories of techniques used in the alpha model. The fundamental analysis looks at macroeconomic factors such as the overall economic and industrial conditions as well as company specific factors. The technical analysis approach uses charting techniques and statistics to study asset trends. The

quantitative analysis applies advanced computational metrics such as those used in physics and machine learning to capture and predict future asset price movements (Treleaven et al. [2013]).

Technical analysis is one of the principle techniques of Alpha model component in Pre trade analysis (Treleaven et al. [2013]). In technical analysis algorithms are employed to predict the future price moments of the assets based on price histories and other trade related information. Theoretically the technical analysis assumes that the market price of the asset reflects all relevant information hence it analyses the price movement patterns rather than examining the underlying factors affecting them (Nuti et al. [2011]).

To summarize, the part of a algorithmic trading system which performs pre-trade analysis based on technical analysis is called the Alpha model. The model generates signals which help the algorithmic trading system make decisions on which assets to trade in. The models used in technical analysis generate signals that identify when to enter and exit a position on a financial asset. These signals are based on the premise that the asset prices move in trends. So when a trend is identified by the algorithm it generates an entry signal and an exit signal when the trend ends. The use of indicators such as moving averages, relative strength index, and stochastic oscillators have been used in the past. Newer techniques involve the use of statistical times series forecasting and more sophisticated machine learning techniques (Nuti et al. [2011]).

LITERATURE REVIEW

There has been a lot of debate in the financial world regarding the effectiveness of Technical Analysis (TA) over its reliability to predict future price movement. The practitioners actively use TA based methodologies in trading however the academia is highly sceptical. This has lead to considerable research on the subject. Park and Irwin have performed an exhaustive analysis on the past and present studies performed on the profitability of technical analysis. In this chapter, only a few notable theoretical and empirical studies are shared which are in line with the general theme of this dissertation.

2.1 THEORETICAL LITERATURE

In this section, we present theoretical studies in support and against technical analysis using historical equity prices. The notable theory against technical analysis is the Efficient Market Hypothesis which is presented with alternative theories that give an explanation of why technical analysis can work in the short term.

2.1.1 THE EFFICIENT MARKET HYPOTHESIS

The first theoretical opponent to the use of analysing past prices to predict the future price movement is the Efficient Market Hypothesis proposed by Fama. The theory of Efficient Market Hypothesis (EMH) claims that “the current stock prices fully reflect all security market information, including the historical sequences of prices, rates of return, trading volume data, and other market-generated information, such as off-lot transactions and transactions by market-makers” ([Brown, 2012, p40]). Therefore the stock prices follow a random walk, and that any profits derived from

timing the market are due entirely to chance. If the markets were not predictable then the most appropriate strategy proposed by the supported of EMH is to buy and hold.

The EMH has three levels of strength based on the assumptions about what information is reflected in the stock prices: weak, semi-strong and strong. Each successive higher efficient form of the hypothesis assumes the inclusion the previous.

- **Weak Form:** A market is efficient in the weak form, if it is able to quickly incorporate the information held in the past prices in the current equilibrium price of the asset. Under this form of market efficiency, the historical trading data cannot be used to gain excess returns on the stock market. This would make technical analysis useless.
- **Semi-Strong Form:** A market would be deemed efficient in the Semi-Strong form of the EMH, if all publically available information is already reflected in the equilibrium price of the asset. At this level as suggested before it is assumed that the market is efficient in the weak form as well. Therefore fundamental analysis would be useless to make extra ordinary returns in the semi strong efficient market.
- **Strong Form:** In the Strong Form the market would exhibit all information including inside information in the equilibrium price of the asset. Therefore in the strong form an investor would not be able to gain consistent gains in the market even if it held non public information.

The implication here is that TA should not be able to make profits by exploiting publicly recorded price histories (Purmonen and Griffin [2015]). Two theoretical models describing the price development in efficient markets are the Random Walk Model and the Martingale or Fair Game Model.

The Martingale model (Fair Game Model) which states that the best estimate for tomorrow's price given the past prices of the asset is today's price. Using a discrete time representation, a martingale is a stochastic process (a random sequence) that the expectation of the next value given the previous values is the last observed value. This mathematically put for price P_t in time t ,

$$E(P_{t+1}|P_t, P_{t-1}, \dots P_{t-n}) = P_t \quad (2.1)$$

Due to this linearity in the price development the expectation for the return is,

$$E(P_{t+1} - P_t | P_t, P_{t-1}, \dots) = 0 \quad (2.2)$$

This implies that the return of the asset based on the past history of prices is zero. The martingale model however doesn't say that the price changes are independent. Further more, Fama described a sub-martingale model in which the expected value of the price at time $t+1$ were greater than or equal to the current period t 's price given the all the information at time t . This implies that the returns at $t+1$ can be greater than or equal to zero. According to Park and Irwin, "an important implication of the sub-martingale model is that no trading rules based only on the current information set can have greater expected returns than ones obtained following a buy and hold strategy in a future period".

The Random Walk model is based on a paper written by Bachelier. The basic idea behind the model is that the market prices move due to a large number of random effects which includes the whims of individual participants in the market and hence these movements can be modelled as Brownian motion (Martinelli [2006]). Mathematically written as,

$$P_t = \mu + P_{t-1} + \epsilon_t \sim IID(0, \sigma^2) \quad (2.3)$$

where μ is the expected price change and ϵ_t is an error term which is independently identically distribution with a mean of zero and a known variance. There are other forms of this model with variations on the error term, for more information please see Campbell et al. [1998]. For market efficiency the random walk model is not required however it is a model that quantitative analyst use to model asset prices (Park and Irwin). Implication of this to technical analysis is that the asset price in the future is random and hence cannot be predicted.

To conclude according to EMH the prices reflect all available information, and hence the speculative markets have no memory. As a results, any analysis on the past price movements will not generate a profit on average. The best an investor can do is to buy and hold an asset to get maximum returns.

2.1.2 ALTERNATIVE MODELS

In recent years many theoretical studies based on behavioural finance have been performed in support of the use of Technical Analysis(TA). Two important theoretical explanations which support the use of TA are Noise Rational Expectation and Noise Traders and Feedback. These models rely on the notion of a noise trader whose trading decisions are irrational and erratic. They postulate that prices adjust slowly to new information due to noise, market frictions, and investor sentiments. Hence these models claim that there exists profitable trading opportunities that exist which can be exploited by the use of technical analysis.

The Noisy Rational Expectations Models (NRE) were developed on the idea that there is asymmetric information in the market among participants. The models was developed by specifying two groups of people: a large group of informed traders and a small group of ill-informed traders. The informed traders are further divided into two group. One group looks for specific market information ahead of the others while the other group looks for advanced indications of future behaviour of the markets. As there are many different types of information which affect asset prices, therefore prices tend to change gradually and frequently. This is contrary to the belief held by the Efficient Market Hypothesis which believes that all participants in the markets are aware of all publicly available informations and the price is updated instantaneously. Based on the theory above, the proponents of NRE believe that the asset prices show a pattern of systematic and slow adjustment of newly available information. This gradual change in prices can result in a very short term predictability of the asset price changes, hence implying an existing of profitable trading opportunities using price analysis(Park and Irwin [2007]).

Drawing from psychology and sociology, the behavioural finance proponents describe the workings of two types of investors, arbitrageurs and noise traders, to develop the model of noise traders and feedback. The market participants called arbitrageurs are the investors who form fully rational trading decisions based on gathered information while the other participants called noise traders based their decisions on technical analysis, irrational beliefs, or sentiments. The noises trader buys assets when the price rises and sells when the price falls. These traders follow positive feedback strategies such that when the price increases further because of the demand from the noisy trader they acquire more assets. The arbitrageurs who are rational see that at some point the assets are overpriced in comparison to their fundamentals, hence sell short. Therefore the price in the long

run comes back to the fundamental level however in the short run it is amplified by the affect of the noise traders. As similar situation arises when the stock falls. This behaviour suggests that profits from technical analysis exists in the short term however they adjust to their fundamental levels in the long run(Park and Irwin [2007]).

2.2 EMPIRICAL LITERATURE

Previously performed studies on technical analysis have mixed results. Presenting details from all the studies performed on the equity markets using technical analysis is out of the scope of this thesis. The aim of this section is to present research which have studied technical analysis on equity markets. There are numerous studies which have evaluated equity and other markets in the long term, however studies on intraday technical analysis is hard to find. A lot of the issues which imply to long term technical analysis also apply to short term. From evaluating past studies the aim of this section is to see how the performance of the mathematical models has been evaluated. In specific, looking at the treatment of the transaction cots, risk, data snooping problems, parameter optimization, out of sample verification, and statistical test adopted for the evaluation of the returns produced by the technical analysis models.

2.2.1 TECHNICAL ANALYSIS ON INTRADAY EQUITY MARKETS

Study performed by Marshall et al. [2008] is related to the study presented in this thesis. The study looks at intraday technical analysis in the US equity markets. The authors of the study claim that the survey of market participants indicate that more emphasis is placed on technical analysis than fundamental especially in the shorter time horizons. They evaluate their technical models on intra day returns based data available from Reuters database over a two year period with a five minute interval. Their results demonstrated that technical analysis was not profitable on their selected data series however there was evidence that a few of the rules are profitable. After adjusting for data snooping bias, none of the rules produced statistically significant results. Following an examination of the study is presented.

The data used in the study considered the price series for the full 2 years and was stitched together to form on time series. This contained the full two years of data. Two sub sets where

also created with each set containing one year worth of data. The basic moments of the data were recorded including the Kolmogorov-Smirnov (D-stat) test for normality and test for autocorrelation. They found that the data for non of the periods was normal and significant autocorrelation for one lag was present. However no tests were performed to check if there data showed any signs of random walk.

Five family of technical rules were tested: standard filter, moving averages, support resistance, channel breakouts, and on balance volume. These wide range of rules were tested so that the data snooping bias can be mitigated. The various rules were considered to avoid the risk that any given rule's profitability is due to chance. The profitability of these rules was tested on two different approaches: White's Reality Check bootstrapping technique and the bootstrapping technique proposed by Brock et al.. Marshall et al. [2008] claims that traditionally technical rules were often based on simple t-test methodology which assumes that the returns are normally distributed, stationary, and independently distributed. However the statistical tests on the data showed that it was not normally distributed. Hence to mitigate this and alleviate concerns of the data snooping bias the bootstrapping methodologies were employed.

The study was a pioneering work which looked into technical analysis on equity market intra day. They also testing two main bootstrapping techniques mentioned above to adjust for data snooping bias and to statistically verify their results. However they didn't account for risk, transaction costs, or how they optimized the parameters used in the study. Also they failed to test if bootstrapping from the original data preserved the structure of the original data set i.e. the auto correlation. The main purpose of the momentum based methodologies is to exploit the structure in the price series. The next section will detail how other researchers have dealt with the issues raised here.

2.2.2 EVALUATION OF TECHNICAL ANALYSIS

In the academic literature most empirical studies have focused on testing the profitability of long term technical trading rules. These studies show evidence that some of the technical trading rules are profitable for the given securities they investigated. The focus of this study are short term trading rules however the testing procedures used in these studies are important to note. Despite their claims these studies are subject to various problems in their testing procedures which are detailed here.

Majority of the studies, (Brock et al. [1992]; Shynkevich [2012]; Tian et al. [2002]; Bessembinder and Chan [1998]), have used long term technical analysis on stock market indices. Indices are not trade-able assets. They provide an average of mixed basket of carefully selected stocks to create a market benchmark. Also Bessembinder and Chan states that “changes in the stock price index understate the actual returns due to the omission of dividends”. Technical trading signals therefore cannot be implemented without purchasing the components of the index in correct proportions. These tests of technical analysis trading rules on index data can be biased due to non synchronous trading (Marshall et al. [2008]).

Risk adjusted returns are less likely to challenge market efficiency. The risk adjustment of the returns is important when looking into the futures markets. Margin is collected by the clearing house (central counterpart) to mitigate credit risk of the position. However, in equity markets, long positions do not require margin to be settled. There can be charges associated with a short position in an equity especially if they are held for a longer period of time. This is not relevant for short intra day positions and Marshall et al. [2008] doesn’t account for any in their study. Riskiness of a position and the associated return can also be used as a comparison between to different returns. When comparing trading rules with benchmark returns, risk can be incorporated into the difference of returns. Lukac et al. use Jensen’s α measured by the CAPM model to determine the significance of the returns after being adjust for risk however the Sharp Ratio is also a good measure.

Lukac et al. [1988] make valid points regarding transaction costs in their study. They claim that trading profits from financial anomalies disappear after adjusting for transaction costs. Two kinds of transactions costs are indicated by this statement: commissions and slippage. Slippage is the difference between the price at which the buy/sell order is executed and the price at which the order is filled by the exchange. There could be a significant difference between the execution price and the price at which the order was filled. The commissions refer to the cost the broker charges the client for direct access to the equity markets. The charges for executing an order by the broker depends on the total volume throughput of the transaction. According to Lukac et al. [1988] round trip price for executing a trade on NASDAQ can be as much as 1.34% which includes both the slippage and the commissions. For intraday high frequency traders this amount represents a very high number. The participants using algorithmic trading systems can come in out of positions in

milliseconds. If the cost of execution is as high as 1.34% per transaction then intraday trading would become prohibitive.

Kaufman states that in the past the cost of equity trade execution were about 1%. After many years of negotiations and the reduction in the cost of electronic infrastructure and maintenance there is now a flat fee of \$9 dollars per 1000 shares round trip. This is about 0.9¢ per share. The commissions for equities favours those with larger positions and the stated price is an estimate. In practice the fees are much lower. However caution needs to be taken as “there is equal damage in overstating costs as there is in underestimating them. By burdening a system with unrealistic fees, test may show a loss instead of a profit, causing it to reject a successful trading method” ([Kaufman, 2013, p.22]).

Apart from trading costs it is also important to appropriately backtest a technical trading methodology. Backtesting allows the researcher to confirm and validate their ideas. “Testing is misguided when it is used to discover a successful trading method by massive scanning of combinations of techniques” ([Kaufman, 2013, 42]). The scanning is not limited to the technical trading rules but also to the parameters of the technical trading systems that need to be adjusted to for each type of asset. For example, Lukac et al. [1988] performed optimization on their training set using select systems which were popularly known at that time to be profitable. The term used generically for this ‘misguided testing’ in academia is called, data snooping bias. According to Marshall et al., if rules are tailored to the data series in question and are profitable because of this tailoring then they will not be profitable in an out of sample forecast. Out of sample forecasting is an important indicator of validation of the techniques presented by authors in their research.

A way to mitigate the data snooping bias is to divide the sample data into two groups: training and validation. It is recommended that a validation procedure need to be put in place where the best performing trading model or models are identified. These models are then checked on the validation sample (Jensen [1967]). This is inline with the methodology used by traders. They tend to develop their rules via backtesting over historical data, then use their performant rules on new unseen data. Lukac et al. studied 12 technical trading systems, and conducted out of sample validation of the optimized trading rules. Statistical signification tests were used to measure the performance of the trading rules.

Most early studies did not conduct robust statistical test to see if the returns from technical

trading were significant. Some studies measure statistical significance using z or t tests under the assumption that the trading rules returns are normally distributed. Applying such conventional statistical tests to trading rule returns is invalid since the distributions of the returns under the null hypothesis of an efficient market is not known (Stephen [1985]). However, Lukac et al. performed normality tests on their returns and were not able to reject the null hypothesis which stated that the returns were normal. The t-test for portfolio return therefore could be an appropriate procedure. A one or two tailed test can be performed. The two tailed t-test test where the null hypothesis states that the gross returns generated from technical trading are zero, or a one tailed test could be conducted to test the statistical significance of the net returns. An alternative strategy for testing the statistical significance of trading profits is the use of a model-based bootstrap methodology. This is generally used when the financial returns are not normal i.e. leptokurtic, autocorrelated, or non stationary (Marshall et al.).

DATA AND EMPIRICAL METHODOLOGY

This chapter presents the data of the study and the technical analysis algorithms. Statistical tests and charts created from the data are shared and analysed. In Section 3.1 the mechanism to run the model profitability tests is presented. Data and associated statistical results are presented in Section 3.2 and this chapter is concluded with a description of the technical analysis algorithms.

3.1 TESTING MECHANISM

The aim of this study is to test the intraday predictability of the chosen technical analysis algorithms on highly liquid stocks. Each stock will be considered separately and each positions long/short will be taken during the day by the trading rules and closed out at the end of the day. The profitability will be calculated for each day on each stock. This is contrary to the strategy used by Marshall et al., where they stitched the days together to create one series for the whole year. Significant movement of stock prices can occur after the markets close. This movement is a cause of risk for an intra day algorithmic trading system hence the position is closed out at the end of the day in this study and the gains or losses are calculated.

The data collected for this study is divided into three parts: training, validation, and testing. This mitigates concerns for data snooping bias. Dedicated real data in the training set is used to optimize the technical analysis algorithms. The optimized subset of the algorithms are then used on the validation set to confirm their performance and to make sure that the chosen parameters have not been over optimized for the training set. Once this is confirmed the optimized algorithms are then tested once on the final test set. The results from the test set are shared in Chapter 4.

Trading costs both the slippage and brokerage fees are considered. The slippage will be accounted

for in a unique way. When a buy signal is generated from a trading algorithm the systems registers a position based on the next available time ticks which is taken as the open price for the next time interval. This is a realistic assumption and the sale of a the asset is made on the last price the next time tick. The algorithms are assumed to take position for each time tick and the return is calculated as the difference between the open and last price of the minute. For the purpose of this study the broker fees can be assumed to be negligible however we do present results with 0.9¢ per round trip adjustment to the aggregated profits. Most algorithmic systems trading intra day are working on high volumes hence the broker fees incurred are very low as discussed in Section 2.2.2 and negligible at very high volumes.

The technical trading rules used in this study are presented Section 3.3. These rules will be compared against a buy and hold strategy. The buy and hold will be calculated based on the opening asset price at the start of the trading and the closing price at the end of each trading day. It is assumed that the trader is able to achieve these prices in practice.

Risk of the positions is not accounted for in this study for two main reasons. The trading algorithms are being tested intraday with a short time period. The associated risk for either a long or a short for equity markets at such short time frames is negligible. A margin call for a short sale can be made by the broker however this is unlikely for intraday trading. Also there is no position at the end of each trading day so the likelihood of cost incurred due to short selling is even lower. However Sharp Ratio is calculated on the returns from each model. This provides a ranking of the models based on risk and return.

The returns from each algorithm are calculated for each stock on the intraday trading data of the final unseen test set. Statistical tests are performed to check the statistical significance of these returns using a bootstrapping methodology as the training set returns examined in the Section 3.2 show that these intraday returns are not normally distributed.

3.2 DATA EXAMINATION

Daily intraday data for five large cap stocks listed on the London Stock Exchange were chosen for this study. These stocks are were picked at random from the constituent stocks of the FTSE100 Stock Index.

Company Name	Ticker	Sector
BHP Billiton plc	BLT	Mining
Diageo plc	DGE	Beverages
GlaxoSmithKline plc	GSK	Pharmaceuticals
HSBC Holdings plc	HSBA	Banking
Tesco PLC	TSCO	Super Market

Table 3.1: Randomly Selected Stocks

Data with one minute interval was collected for each of these stocks for 27 days from July 20th, 2015 to August 25th, 2015 excluding public holidays. The intra day data is available from 8:00 am in the morning to 4:30 pm making a total of approximately 500 data points. Google Finance was used to source the data through their API (application programming interface) which allows for convenient extraction to csv files. For each intra day the one minute time tick data available from Google Finance contains open, high, low, close, and volume. The close prices for each tick are the price for the last trade. This data is divided into three sets as mentioned above: training, validation, and test sets.

- **Training Set:** Consists of 13 days from 20/07/2015 - 05/08/2015 excluding holidays. This data set is used to optimize the parameters of the algorithms and choose the most performant. Statistical analysis is performed to get an understanding of the intraday training data which is discussed in Section 3.2.1.
- **Validation Set:** Consists of 7 days from 06/08/2015 - 14/08/2015 excluding holidays. The validation set is used to validate the algorithms and their selected optimized parameters. An iterative process of on the training and validation set was done to arrive at the final set of algorithms. However no analysis is performed on validation set so that this data can be used as a proxy for the final unseen test set.
- **Test Set:** Consists of 7 days from 17/08/2015 - 25/08/2015 excluding holidays. This set is the unseen final test set. No analysis was performed on this set prior to running the final test results in order to eliminate any data snooping bias. Only select algorithms are run on this set and the results are discussed in the chapter 4.

3.2.1 DESCRIPTIVE STATISTICS

In this section, the data for the training set is analysed. One minute returns are calculated on the data set by taking the difference between the opening price and the closing price for the minute. Following is the definition:

$$r_t = P_{close,t} - P_{open,t} \quad (3.1)$$

The reason for choosing the difference between the opening and closing price for the minute is to account for slippage. In intraday trading many thousand of stocks change hands between a minute. The price moves after the last closing price has been conveyed to the market participant. The algorithmic trading system employed by the market participant will make a trading decision based on the closing price but will not achieve this price in settlement. Hence a good estimate of the settlement price is the opening price for the next time tick.

The algorithmic trading system can take a position every minute depending on the direction of the movement it perceives the asset will take. This is assumed by the study. Hence the returns are calculated of the price movement within each minute of intraday trading.

The statistics calculated on the training data set for all the five stocks are presented in the Table A.1 in the Appendix A. The kurtosis and skewness for all stocks are significant at the 5% level. The return distribution is highly leptokurtic and indicate the presence of fat tails in each of the stock return distributions. The Kolmogorov-Smirnov test was used to test for normality as used by Marshall et al.. The D statistic (labelled D-Stat) returned by the test rejects normality for all the stocks with 1% significance. The Jarque Bera test statistic was also calculated however the values were very large hence it is not reported in the table.

Following the procedure of Chen, the returns of the stocks are testing for the violation of the random walk property by having position autocorrelated statistically significant returns. The test was conducted allowing for a non-zero drift in the random walk. The variance ratio statistics for various lags that are reported in Table A.1 in the Appendix A. All of the lags suggest a strong rejection of the null hypothesis which states that the returns are a pure random walk. The heavily negative values can suggest autocorrelation between the returns or high levels of heteroskedasticity in the dataset. The Durbin Watson test statistic for presence of autocorrelation is around two for all stocks which rejects the hypothesis that there is any autocorrelation present.

To visualize the return data for each stock the following items are generated: visual plots, normalized histogram, autocorrelation function plots (ACF), the partial autocorrelation function plot (PACF), and the Quantile-Quantile (QQ) plot against the normal distribution. These can be seen in figures A.1, A.2, A.3, A.4, and A.5 in the Appendix A. Please see the corresponding numbers up to 5 lags in the table A.1 as well. The PACF and the ACF charts show that there is no significant autocorrelation present in the returns.

In the figures A.1 through A.5 the plot of the normalized log returns shows high levels of variation. There are many returns more than 3 standard deviations from the mean. The plot of the histograms shows the extreme leptokurtosis with very long tails as is expected from the statistical measures. The QQ plots verify the same. The QQ plots also show that the data points are close to normal between approximately $\{-2, 2\}$ standard deviations and move away beyond these bounds. These fat tails of the distributions support the plot graphs of the returns that there are unusually large number directional moves that are longer than would be expected if the prices were randomly distributed Kaufman [2013]. This supports the rejection of the random walk properties of these returns as the variance ratio tests indicated. According to Kaufman, fat tails demonstrate the existence of trend profits which can be exploited by trend systems.

To conclude the statistical analysis of the training set returns it can be said that the returns don't exhibit random walk properties and are not normally distributed. The data distribution is highly leptokurtic with very large tails. There are quite a few extreme outliers beyond 10 standard deviations. It is these outliers that the algorithm using technical analysis aims to use to make excess gains.

3.3 TECHNICAL TRADING RULES

In this section, the details of the technical trading rules which are the focus of this study will be explained. The mechanics used to predict the future stock move and how to indicate a buy or sell signal will be discussed. The parameters for each of the algorithms will be determined by optimizing their performance on the training and validation data sets. Parameters for each stock will be individually determined and used on the final unseen test set.

3.3.1 NAIVE FILTERS

Two simple filter rules as described by [Schmidt, 2011, p.105] are used. This first (SIMPLE1) is a very naive trading rule, where the system will take long (short) position at the next time step if in the previous time step the closing price of the stock P_t was higher(lower) than the former closing price P_{t-1} by a given threshold $\delta > 0$. The signal is generated based on the following equation:

$$\begin{aligned} \text{Buy} : P_t/P_{t-1} &> 1 + \delta \\ \text{Sell} : P_t/P_{t-1} &< 1 - \delta \end{aligned} \tag{3.2}$$

It is recommended that the δ be chosen to be higher than the assets daily volatility. The other (SIMPLE2) described by [Schmidt, 2011, p.105] is where a long(short) position is taken if the current time period price P_t is greater(less) than the maximum(minimum) of past k period values. The signal is generated as follows:

$$\begin{aligned} \text{Buy} : \frac{P_t}{M(k)_{high}} &> 1 + \delta \\ \text{Sell} : \frac{P_t}{m(k)_{low}} &< 1 - \delta \end{aligned} \tag{3.3}$$

where $M(k)_{high} = \max(P_{t-1}, \dots, P_{t-k})$ and $m(k)_{low} = \min(P_{t-1}, \dots, P_{t-k})$. The δ for both SIMPLE1 and SIMPLE2 will be chosen by optimizing their performance on the training set data and validating it on the validation set.

3.3.2 SIMPLE MOVING AVERAGE FILTERS

Moving Averages are the most commonly used technical analysis technique. The simple moving average (SMA) algorithm compares two different moving averages a fast and a slow one to make trading decisions. A simple moving average is defined as follows:

$$sma(P, k) = \frac{P_{t-1} + P_{t-2} + \dots + P_{t-k}}{k} \tag{3.4}$$

To make a decision on when to trade the system compares the relative difference or return between the Fast(k is small i.e. 2) and Slow (k is big i.e 20) moving averages. An example of three simple

moving average filters on Diageo plc stock (Ticker:DGE) can be seen in Figure A.6 in the Appendix A¹

$$r_{P,t} = \frac{sma(P, FAST) - sma(P, SLOW)}{sma(P, FAST)} \quad (3.5)$$

The signal from the moving average based algorithm is as shown in the equation 3.6. The parameters which must be chosen when using simple moving average filter is are the window for the fast and the slow moving averages and the delta to generate the signal. These parameters will be selected using the training and validation data sets for each stock.

$$\begin{aligned} Buy : r_{P,t} &> \delta \\ Sell : r_{P,t} &< \delta \end{aligned} \quad (3.6)$$

3.3.3 EXPONENTIAL MOVING AVERAGE

The simple moving averages in the signal generation given above can be replaced by weighted moving averages. Simple moving averages weights each of the past prices with an equal weight. If there is autocorrelation present between the prices usually there is more correlation between the first few lags. Hence the lags closer to the current stock price should have more weighted.

The mechanism to weight a price series is given in equation 3.7. The weighted moving average at time t is the average of the previous n prices with each of the past prices having its own weighting factor w_i . These weights chosen for this study will be exponential.

$$W_{t,k} = \frac{w_t P_t + \dots + w_{1-k} P_{t-k}}{w_t + \dots + w_{t-k}} \quad (3.7)$$

The signal generation for the exponential moving average (EMA) remains the same as first we calculate the return as per 3.5 and then the signal is generated using 3.6, and also the parameter selection process.

¹The diagram has been produced using Python Matplot library <http://matplotlib.org/>

3.3.4 MOVING AVERAGE CONVERGENCE/DIVERGENCE

Moving average convergence divergence (MACD) is a type of momentum technical indicator. Momentum indicators use the rate of change of the price as mechanism to smooth prices. The momentum for k days is calculated as $M_t = P_t - P_{t-k}$. Schmidt, p.111 describes MACD as a widely popular momentum indicator among practitioners, where a difference between a fast and slow exponential moving average is used. The equation 3.8 details the formula.

$$MACD_t = ema(P_t, 12) - ema(P_t, 26) \quad (3.8)$$

The signal is generated by further smoothing the $MACD_T$ line as given in equation 3.9. The buy signal is generated when the signal line crosses the MACD line from below to above, and the sell signal is generated when the signal line crosses the MACD line from above to below.

$$signal = ema(MACD_t, 9) \quad (3.9)$$

An example of the $MACD(12, 26, 9)$ is shown in the bottom of the figure A.6 in the Appendix A.

3.3.5 RELATIVE STRENGTH INDEX

The MACD described in the previous section can also be considered an oscillator patterned technical indicator as the lines oscillate around zero. The most popular oscillator according to Schmidt, p.111 is the relative strength index (RSI). This indicator is determined with directional price moves during a given time period k . Usually k is set to 14.

$$RSI_k = \frac{100 \times RS}{1 + RS}, RS = \frac{n_{up}}{n_{down}} \quad (3.10)$$

The n_{up} and n_{down} in RS are the number of upward moves and downward moves of the closing price respectively. Kaufman, p.386 states that typically the threshold values suggesting that a stock is over/under sold is 70/30. For highly volatile stocks this can widen to 80/20. The parameters which can be adjusted in this algorithm are the up/down percentages and the k period.

3.3.6 KALMAN FILTERS

Martinelli and Rhoads present an intriguing alternative often used in engineering for controlling noisy systems. In their article Martinelli and Rhoads [2010], use Kalman filters (KAL) to predict stock market movements. The Kalman Filter “is a two-stage algorithm that assumes there is a smooth trend line within the data that represents the true value of the market item before being perturbed by market noise”(Martinelli and Rhoads [2010]). In the first stage the data is fit to a suitable model. This model is used to predict the next $t+1$ data point and the error in the prediction. In the second stage the prediction is corrected by combining the actual value to create a trend value. These two stages repeated to create a cycle of prediction and correction(Martinelli and Rhoads [2010]).

For this research thesis, the most simplest Kalman Filter was employed. It is assumes that the current price P_t is the best prediction for the next price evolution in time $t+1$. Equation 3.11 shows the calculation of the prediction and the calculation of the estimation error.

$$\begin{aligned} P_{t|t-1} &= P_{t-1|t-1} \\ e_{t|t-1} &= e_{t|t-1} + q_t \end{aligned} \tag{3.11}$$

where e is the estimation error and q is the process error. The process error is unknown and needs to be fitted or assumed. The estimation error is adjusted with every cycle in the update stage of the process. The equations 3.12 show the calculation of the Kalman update.

$$\begin{aligned} P_{t|t} &= P_{t|t-1} + K_t(y_t - P_{t|t-1}) \\ K_t &= e_{t|t-1}(e_{t|t-1} + r)^{-1} \\ e_{t|t} &= (1 - K_t)e_{t|t-1} \end{aligned} \tag{3.12}$$

The value $y_t - P_{t|t-1}$ is called the innovation or the new information which was not already predicted in $P_{t|t-1}$. K the gain decides how much of this new information should be part of the correction of the estimated $P_{t|t}$. For a more in depth understanding of application of Kalman Filters please see Rojas [2003] and Martinelli and Rhoads [2010].

The Kalman indicator will make a naive prediction of the filtered value of the price. The signal is generated by comparing the $P_{t|t-1}$ with δ threshold which is chosen through optimization.

EMPIRICAL FINDINGS

The study has found that the MACD algorithm performed the best on the test data set. The short trades added most to the profitability of the trading algorithms. After examination of the test data charts it is evident that the market is in a bearish state. The models take advantage of this to earn a excess profit against the buy and hold strategy.

4.1 SUMMARY RESULTS

Table 4.1: Summary Results of all Models

Ticker	Profit	SR	Sell	Buy	#Buy	#Sell
MACD	448.27	3.1860	272.22	176.05	8025	9146
EMA	318.17	1.9201	201.41	116.76	3542	13804
SMA	244.55	1.7313	164.6	79.95	2489	14857
RSI	199.84	1.7119	183.04	16.8	1096	11140
KALMAN	155.32	0.6751	135.03	20.29	7479	9692
SIMPLE1	113.51	-0.1198	57.52	55.99	1045	1092
SIMPLE2	110.13	-0.2569	25.29	84.84	2051	2090
BH	81.79	n/a	n/a	n/a	7	7

As per the methodology discussed in the Chapter 3, the models were optimized on the training and validation set. They were then run on the out of sample test set. Table 4.1 presents the results without broker fees. The models are sorted based on their performance in terms of profits generated over all, with the best performing model at the top and the worst at the bottom. The buy and hold strategy (BH) is at the bottom of the table. The SR column gives the Sharp Ratio, which is

a comparison of the return given the risk. Columns 3 and 4 give the breakdown of the profits by buy (long) and sell (short). The last two columns summarize the results for the number of buys and the number of sells made by each of the models.

Table 4.2: Results with Broker Fees 0.9 cents

Ticker	Profit	Trips	Cost	Adj Profit
MACD	448.27	8585	77.27	371.00
EMA	318.17	8673	78.06	240.11
SMA	244.55	8673	78.06	166.49
RSI	199.84	6118	55.06	144.78
SIMPLE1	113.51	1069	9.62	103.89
SIMPLE2	110.13	2070	18.63	91.50
BH	81.79	7	0.06	81.73
KALMAN	155.32	8585	77.27	78.05

The above Table 4.2 show the summary table with hypothetical cost of 0.9¢ per 1000 trades. The Kalman Filter performs worst that the buy and hold making only £78.05 against a buy and hold profit of £81.73. However, MACD still performs at the top of the list.

Table 4.3 provides a breakdown of the results by ticker and model. The stocks have been sort in descending order based on the totals with the top performing stock over all models at the top, and the best performing model in the first row. The MACD outperforms the all the other models on average. However it doesn't perform well on all days. The moving average based models, MACD, SMA, and EMA are all top performers.

Table 4.3: Summary Results of Models vs Ticker

Ticker/Model	MACD	EMA	SMA	RSI	KALMAN	SIMPLE1	SIMPLE2	BH	Total
DGE	125.95	157.37	150.13	127.68	60.97	37.72	51.26	93.29	804.37
HSBA	66.07	67.24	49.00	51.98	36.97	25.73	1.13	46.05	344.17
BLT	167.89	44.76	11.78	-18.07	60.37	19.36	46.32	-18.23	314.18
GSK	71.00	44.70	30.22	37.21	-12.08	27.93	10.84	-28.61	181.21
TSCO	17.36	4.10	3.42	1.04	9.09	2.77	0.58	-10.71	27.65
Total	448.27	318.17	244.55	199.84	155.32	113.51	110.13	81.79	1671.58

4.2 ANALYSIS

A detailed view of the results is provided in the Table A.9 in the Appendix A. One concern is that the returns achieved by the models are much higher than the buy and hold strategy. Also the highest performing model in the table 4.1 MACD is four times as profitable than the lowest performing models SIMPLE2. To explain such high numbers please see the breakdown of the of each model by date for each stock in Table A.9. The 24/08/2015 is always on top as the highest performing day in the amongst all the other days and this is the day where most of the positive profits were made by the algorithms. This happens to be the day when the FTSE100 lost about 5% of its value. A intra day trading model is supposed to help the participant take advantage of high variability in the market and this is an example of that.

Inspection of the statistics for each of the stock provides a few key insights into the ordering shown in Table 4.3 where DGE is most profitable on average with the rest of the stock in descending order of profitability. An analysis of the autocorrelation and partial autocorrelation charts shows that test data of DGE showed signs of significant autocorrelation in comparison to others. The charts for all the stocks for the test data Figures A.7,A.8,A.9,A.10, and A.11 in the Appendix A. This could explain the higher predictability of the models on the respective stocks. The autocorrelation in the lags beyond 3 is indicative of the lags of the models which performed the best. In the summary table A.4 which gives are breakdown of the optimized parameters for each model per stock by model, the fast and slow moving averages used by the MACD model are between 7 and 15 each respectively.

The Sharp Ratio(SR) is a good test of returns given the risk taken. Please note that the Sharp Ratio calculated here is calculated based on arithmetic returns not geometric returns, and the returns are also based on the difference between the opening price and the closing price. In this study, the aggregate SR over the trading days and stocks is presented as a comparison of the results given by each model verses the variability of their returns. In table 4.1 the aggregate sharp ratio is presented for each model. The simple models (SIMPLE1 and SIMPLE2) have negative sharp ratios which is expected as the variability of the returns provided was extremely high because of their naive assumptions. It can be said that in future the simple models could potentially under perform. The MACD provides the best returns with a very high aggregate sharp ratio. Since the returns are not normal as presented in Chapter 3 a simple parametric statistical test cannot be performed.

All the profits presented are significant based on bootstrap test. A bootstrap test was performed to test whether the mean profits were statistically different from zero. The results vector was selected by a IID Bootstrap selection of 999 sets. The critical values based on percentile were selected and the null hypothesis that the mean profit was statistically different from zero was rejected with a 1% level of significance. A more sophisticated bootstrap was not required such as those performed by Brock et al. [1992] because the test data was not used to train the algorithms hence data snooping was not an issue.

On average, for most high performing models the profit from the sell trades is higher than the buy trades. In table ??, EMA, SMA, and RSI have disproportionately higher returns achieved from short trades (sell) than long trades. This is very different to the other studies previously conducted namely Brock et al. [1992] where it was noticed that the long trades contributed the most to the aggregate profit of the model. Their studies were performed on long term stock movements and this could be the discernible difference between the observations in this study and theirs. The study that performed intra day test Marshall et al. [2008] didn't provide any results based on long or short position. None of their models tested provided any significant returns. However it is to be noted that the test set shows a significantly bearish market. This means that short trades would be profitable, and the models by choosing to execute these trades would make significant profits.

CONCLUSION

The question proposed was whether technical analysis algorithms used in the alpha model part of the algorithmic trading system able to generate significant profits in intraday London equity markets. The study found that the Moving Average Convergence and Divergence model performed the best and was able to give statistically significant returns on the data set chosen. The evidence of profitability of the MACD model in the London Market goes against the Efficient Market Hypothesis proposed by Fama and supports the alternative models based on behavioural finance which state that the markets have short term momentum which can be exploited to make statistically significant returns. The study supports the use of similar techniques by practitioners as claimed by Marshall et al. in the short term.

The study is unique in a few respects compared to the others studies performed by Martinelli and Rhoads; Martinelli and Rhoads; Brock et al.; Shynkevich. This study is performed on one minute intraday quotes and uses a unique way to account for slippage. The slippage is a real concern when comparing to real live systems where the price the trade is executed at is not achieved at settlement. To account for the slippage the study used returns based on the open price and close price for each minute. Concerns of data snooping are alleviated by the use of three sample data sets of the intraday prices: training, validation, and test set. Training and validation sets are used to optimize and select performant models, while the unseen test set was used to calculate the results.

Even though, all the profits were tested to be significant using bootstrap methodology, but there wasn't any statistical test performed to check if the results seen between all the different models were statistically significant and if a ranking can be obtained. It is recommended by this study that in future a StepM model which is an advanced statistical technique to rank models can be used to perform such a test. For more information please see Romano and Wolf [2005]. Also future studies

could consider other intraday market information like high,low, and volume which is also available at one minute intervals.

APPENDIX

A.1 ACKNOWLEDGEMENT

I would like to thank and acknowledge the following people for helping me to complete this study.

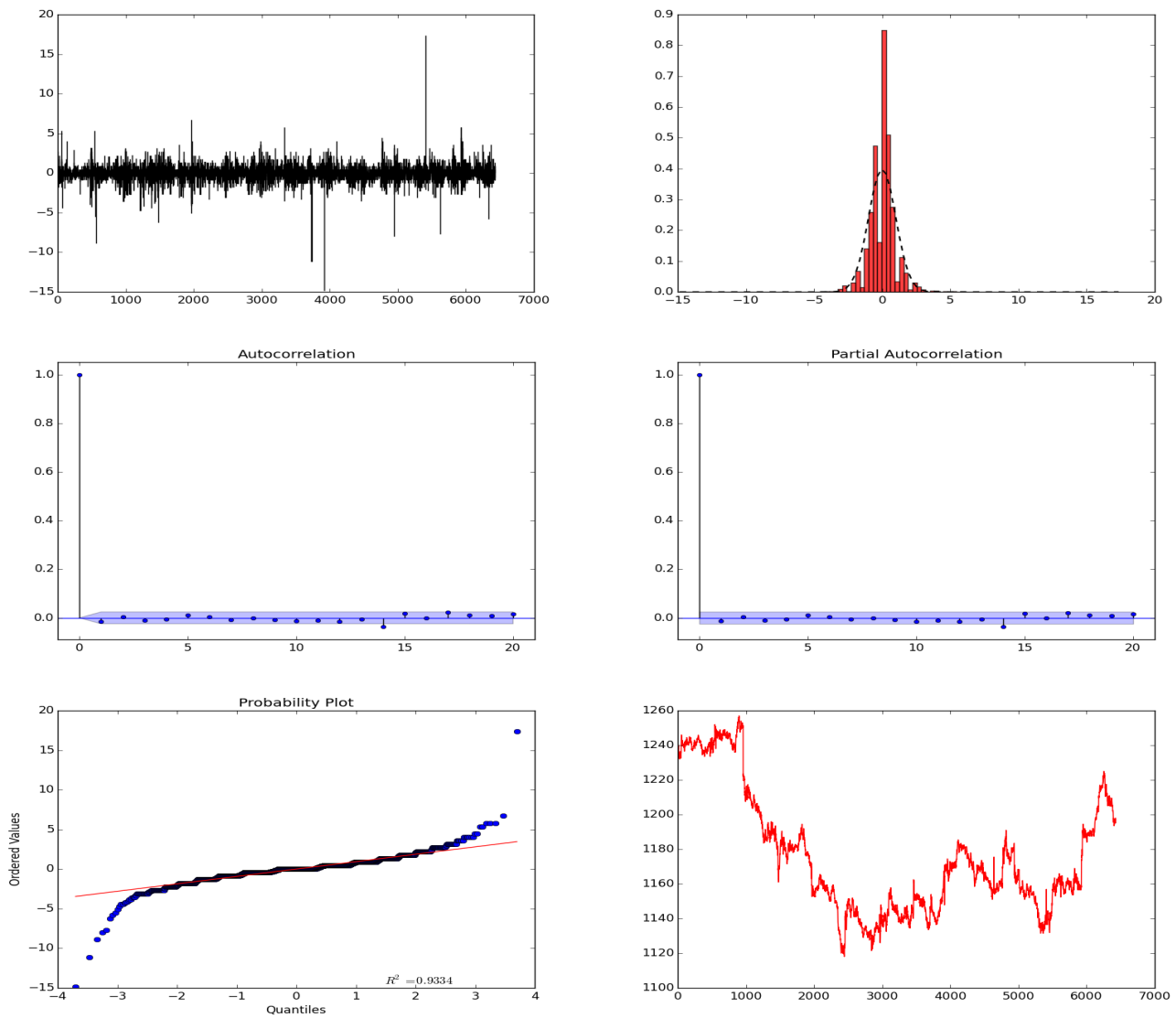
- My supervisor Dr. Xueqi Dong for her help and advice.
- Abderazzak Benkirane and Denato Falce, Quantative Risk at CME Group London for providing key insight into the world of algorithmic trading and suggesting which models are most popular amongst practitioners.
- I would like to thank Dr. Kevin Sheppard for his statistical library in Python and the exceptionally details website with examples on how to use it.
- Dr. Jon Breitenbucher for putting an exceptionally detailed instruction on how to install and use Latex. Especially providing this template which I have used.
- My Partner for the support and care without which doing this study would have been difficult.

A.2 TRAINING DATA STATISTICS AND GRAPHS

Table A.1: Training Set Statistics

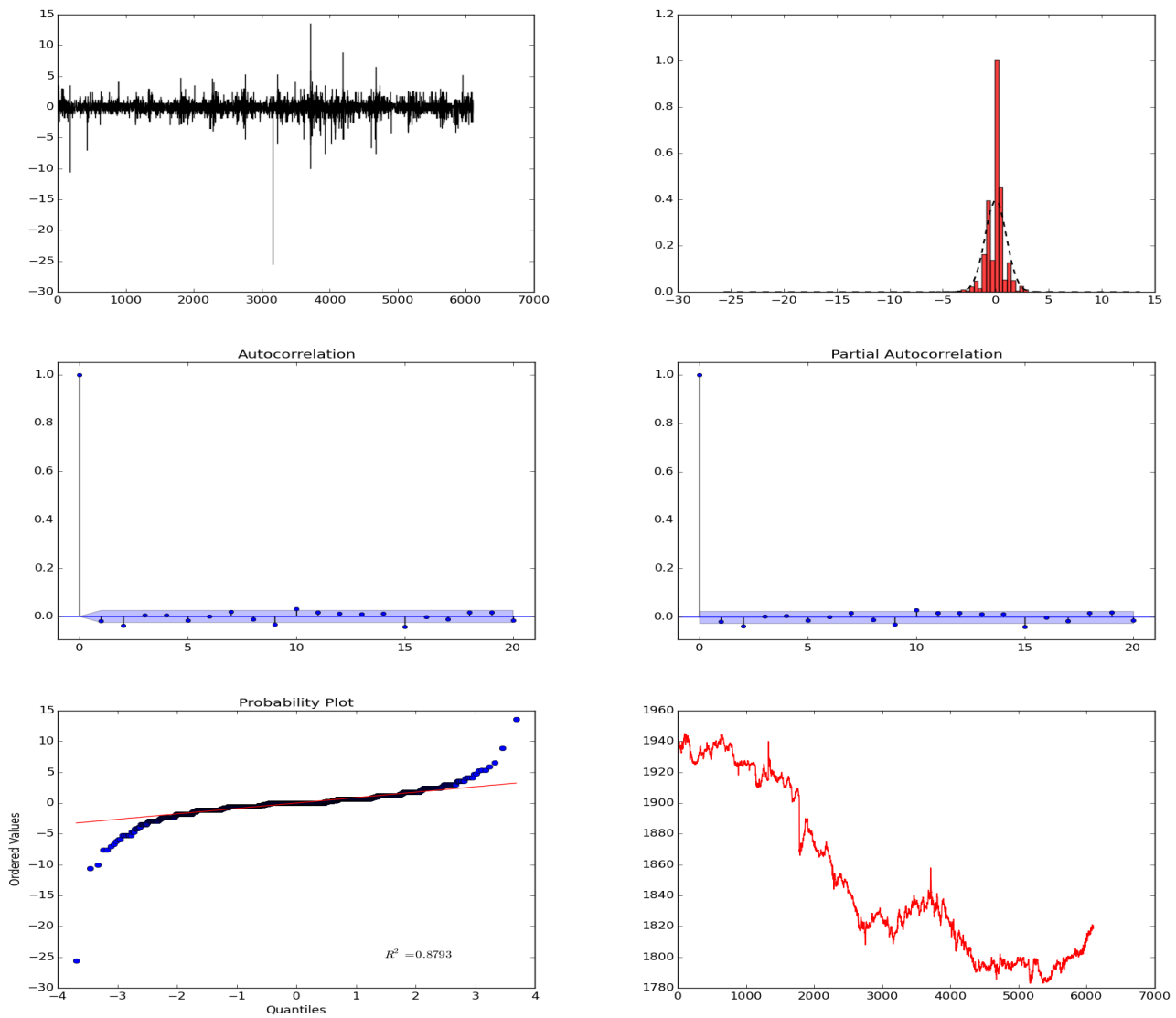
SYMBOL	BLT	DGE	GSK	HSBA	TSCO
MEAN	-0.0166	-0.0278	0.0053	-0.0004	0.0018
STDEV	1.1214	0.8472	0.9944	0.2458	0.1041
KURTOSIS	<i>29</i>	<i>86</i>	<i>710</i>	<i>155</i>	<i>43</i>
SKEW	<i>-0.12</i>	<i>-3.00</i>	<i>-13.10</i>	<i>-4.73</i>	<i>2.51</i>
D-STAT	<i>0.125</i>	<i>0.198</i>	<i>0.220</i>	<i>0.333</i>	<i>0.406</i>
DW	<i>2.027</i>	<i>2.035</i>	<i>1.946</i>	<i>1.993</i>	<i>1.947</i>
ACF1	-0.014	-0.019	0.025	0.002	0.022
ACF2	0.004	-0.038	0.086	-0.006	-0.017
ACF3	-0.010	0.004	-0.010	-0.013	0.034
ACF4	-0.005	0.006	-0.013	0.013	-0.004
ACF5	0.011	-0.015	0.024	-0.011	0.007
PACF1	-0.014	-0.019	0.025	0.002	0.022
PACF2	0.004	-0.038	0.086	-0.006	-0.017
PACF3	-0.010	0.003	-0.014	-0.013	0.035
PACF4	-0.006	0.005	-0.020	0.013	-0.006
PACF5	0.011	-0.015	0.027	-0.011	0.008
VR3	-26.08	-15.50	-5.60	-13.35	-21.62
VR6	-24.38	-14.56	-5.53	-13.06	-19.64
VR9	-22.89	-13.91	-5.48	-12.80	-18.44
VR12	-21.70	-13.51	-5.41	-12.52	-17.52
VR16	-20.42	-12.98	-5.35	-12.20	-16.61
VR24	-18.48	-12.15	-5.23	-11.61	-15.31

- D-Stat: Kolmogrov-Smirnov test D Statistic
- VR: Variance Ration Statistic
- ACF: Autocorrelation Function
- PACF: Partial Autocorrelation Function
- DW: Durbin Watson statistic for autocorrelation

Figure A.1: BLT (BHP Billiton) Training Set

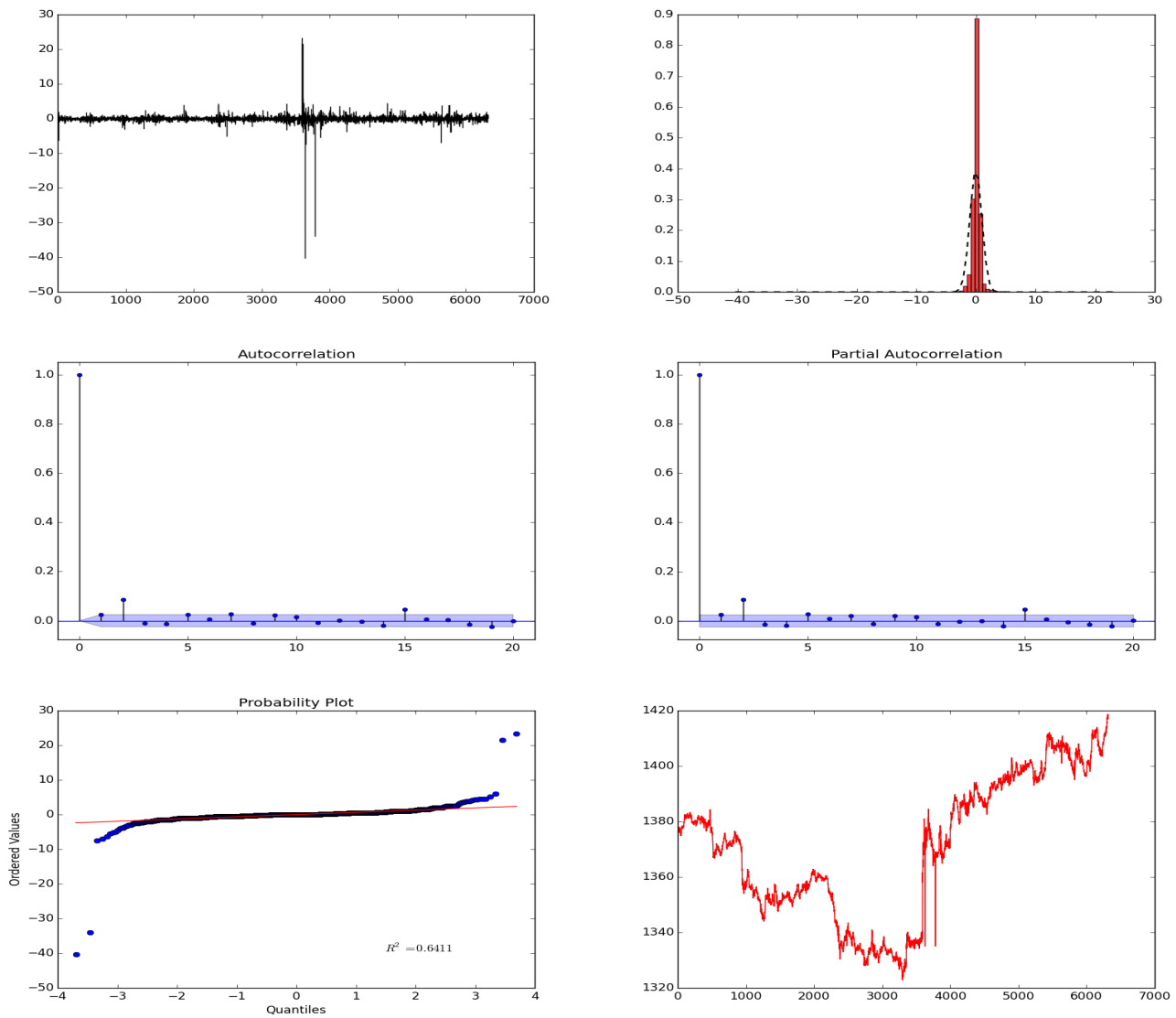
Figures starting from the top left hand corner ...

- Return plot (Open - Close)
- Normalized histogram of the returns
- Autocorrelation function of the returns
- Partial autocorrelation function of the returns
- Quantile-Quantile(QQ) Plot showing the horizontal red line as the normal distribution and the blue dots are the returns plotted against it.
- Simple plot of the closing prices of the Stock

Figure A.2: DGE (Diageo) Training Set

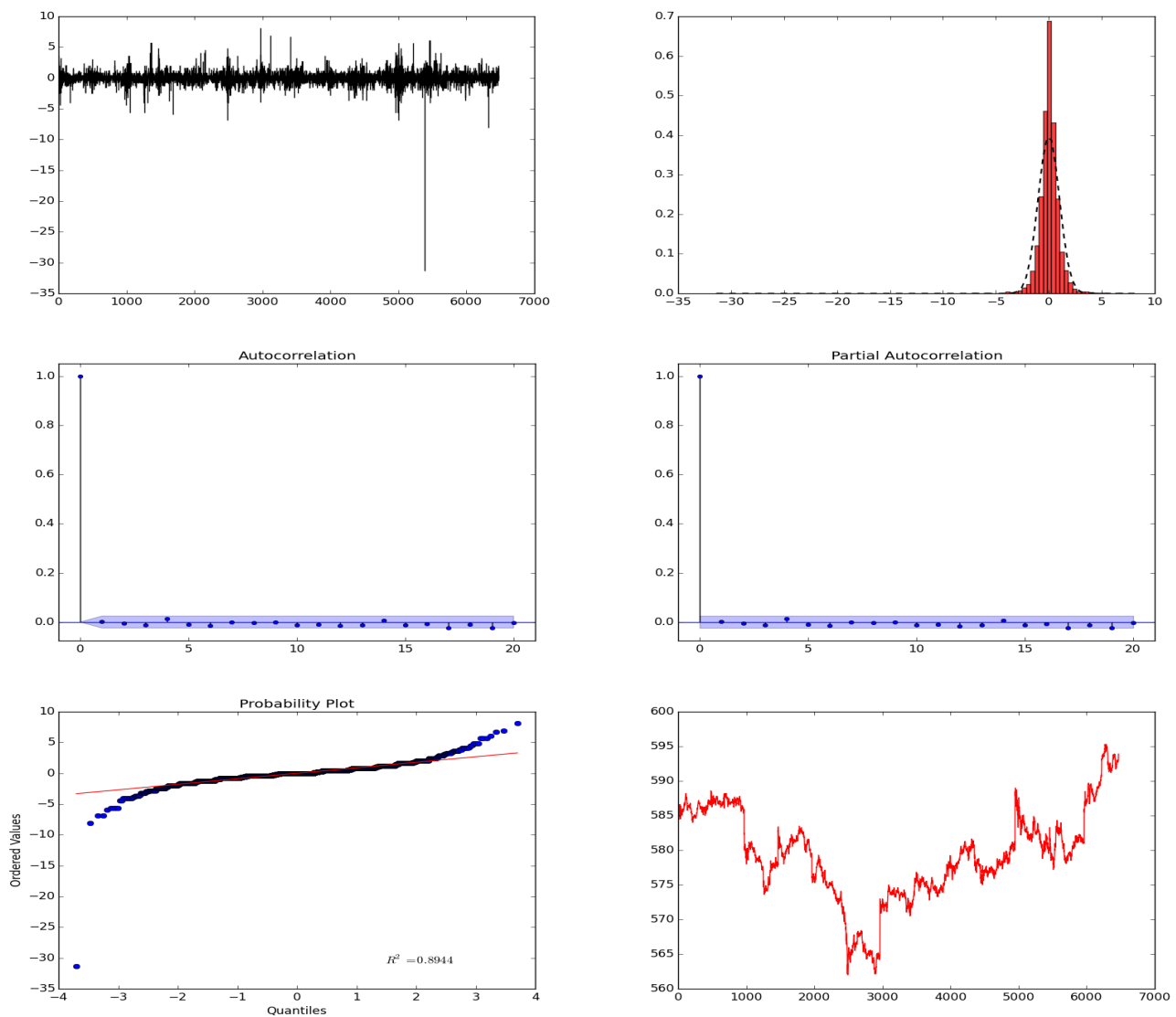
Figures starting from the top left hand corner ...

- Return plot (Open - Close)
- Normalized histogram of the returns
- Autocorrelation function of the returns
- Partial autocorrelation function of the returns
- Quantile-Quantile(QQ) Plot showing the horizontal red line as the normal distribution and the blue dots are the returns plotted against it.
- Simple plot of the closing prices of the Stock

Figure A.3: GSK(GlaxoSmithKline) Training Set

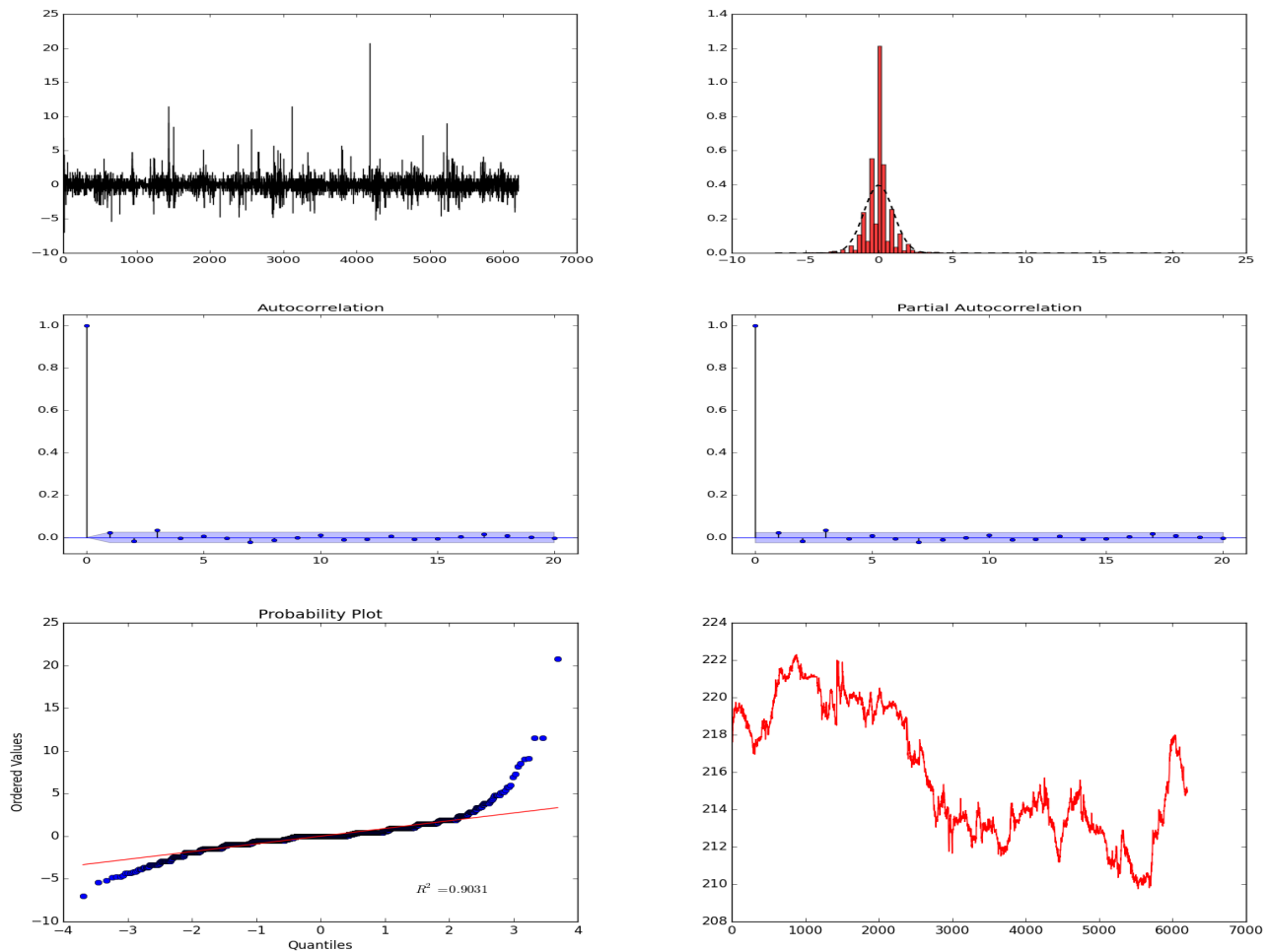
Figures starting from the top left hand corner ...

- Return plot (Open - Close)
- Normalized histogram of the returns
- Autocorrelation function of the returns
- Partial autocorrelation function of the returns
- Quantile-Quantile(QQ) Plot showing the horizontal red line as the normal distribution and the blue dots are the returns plotted against it.
- Simple plot of the closing prices of the Stock

Figure A.4: HSBA (HSBC Holdings) Training Set

Figures starting from the top left hand corner ...

- Return plot (Open - Close)
- Normalized histogram of the returns
- Autocorrelation function of the returns
- Partial autocorrelation function of the returns
- Quantile-Quantile(QQ) Plot showing the horizontal red line as the normal distribution and the blue dots are the returns plotted against it.
- Simple plot of the closing prices of the Stock

Figure A.5: TSCO (Tesco) Training Set

Figures starting from the top left hand corner ...

- Return plot (Open - Close)
- Normalized histogram of the returns
- Autocorrelation function of the returns
- Partial autocorrelation function of the returns
- Quantile-Quantile(QQ) Plot showing the horizontal red line as the normal distribution and the blue dots are the returns plotted against it.
- Simple plot of the closing prices of the Stock

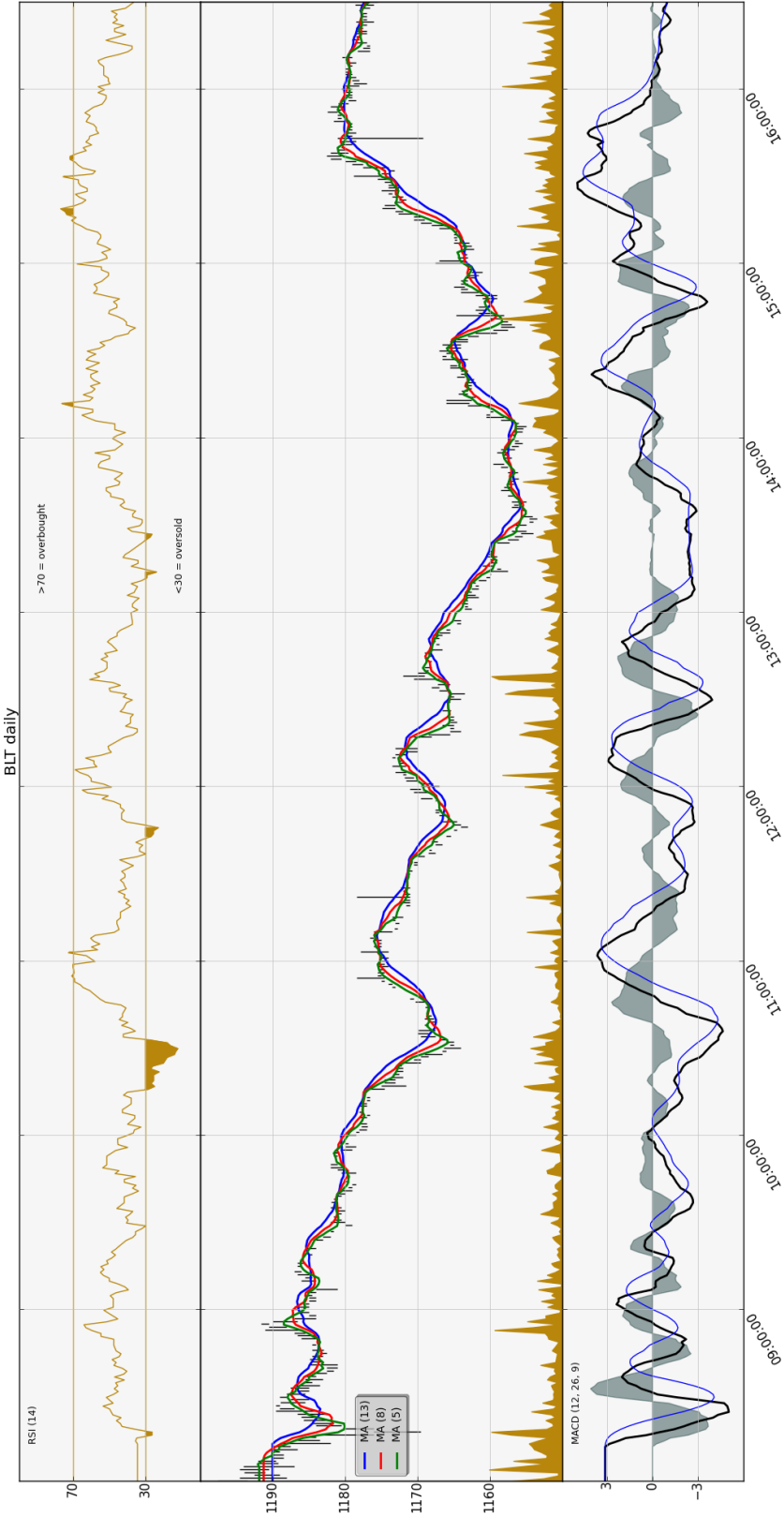


Figure A.6: Technical Indicators DGE Intraday Prices (8/6/2015)

A.3 SUMMARY RESULTS

Table A.2: Summary Result for the EMA Model

Ticker	BH	EMA	SR	Sell	Buy	#Buy	#Sell	Fast	Slow	δ
BLT	-18.23	44.76**	0.1927	7.29	37.47	938	2570	3	7	0.0005
DGE	93.29	157.37**	0.7121	123.83	33.54	545	2873	3	7	0.0005
GSK	-28.61	44.7**	0.0438	14.66	30.04	742	2678	2	6	0.0003
HSBA	46.05	67.24**	0.8410	59.05	8.19	663	2852	2	6	0.0004
TSCO	-10.71	4.1**	0.1301	-3.42	7.52	654	2831	3	7	0.0005

Table A.3: Summary Result for the Kalman Model

Ticker	BH	KAL	SR	Sell	Buy	#Buy	#Sell	Window
BLT	-18.23	60.37**	0.2527	17.85	42.52	1438	2035	5
DGE	93.29	60.97**	-0.0592	84.78	-23.81	1522	1861	5
GSK	-28.61	-12.08**	-0.1843	-13.54	1.46	1461	1931	4
HSBA	46.05	36.97**	0.4360	45.10	-8.13	1483	1997	5
TSCO	-10.71	9.09**	0.2300	0.84	8.25	1575	1868	6

Table A.4: Summary Result for the MACD Model

Ticker	BH	MACD	SR	Sell	Buy	#Buy	#Sell	Fast	Slow
BLT	-18.23	167.89**	0.7295	71.61	96.28	1576	1897	6	13
DGE	93.29	125.95**	0.2450	109.55	16.40	1601	1775	7	15
GSK	-28.61	71**	0.5090	26.79	44.21	1587	1798	6	12
HSBA	46.05	66.07**	0.8927	59.65	6.42	1588	1892	6	12
TSCO	-10.71	17.36**	0.8098	4.62	12.74	1673	1784	5	11

Table A.5: Summary Result for the RSI Model

Ticker	BH	RSI	SR	Sell	Buy	#Buy	#Sell	UP	DOWN	Window
BLT	-18.23	-18.07**	-0.0729	-18.49	0.42	110	2468	63	20	19
DGE	93.29	127.68**	0.8183	125.11	2.57	44	2337	62	15	12
GSK	-28.61	37.21**	0.1551	19.51	17.7	248	1688	65	24	20
HSBA	46.05	51.98**	0.9082	56.79	-4.81	63	2606	63	18	19
TSCO	-10.71	1.04**	-0.0968	0.12	0.92	631	2041	63	21	18

Table A.6: Summary Result for the SIMPLE 1 Model

Ticker	BH	SMPL1	SR	Sell	Buy	#Buy	#Sell	Threshold
BLT	-18.23	19.36**	-0.1805	10.00	9.36	150	147	0.00209
DGE	93.29	37.72**	-0.1304	26.97	10.75	202	226	0.00078
GSK	-28.61	27.93**	0.5767	-3.16	31.09	69	54	0.00142
HSBA	46.05	25.73**	-0.1709	27.09	-1.36	369	439	0.00059
TSCO	-10.71	2.77**	-0.2148	-3.38	6.15	255	226	0.00100

Table A.7: Summary Result for the SIMPLE 2 Model

Ticker	BH	SMPL2	SR	Sell	Buy	#Buy	#Sell	Threshold	Period
BLT	-18.23	46.32**	-0.0888	-16.10	62.42	578	592	0.00057	2
DGE	93.29	51.26**	0.0150	41.76	9.50	271	263	0.00057	2
GSK	-28.61	10.84**	-0.1217	-4.03	14.87	414	429	0.00037	2
HSBA	46.05	1.13**	-0.0574	7.66	-6.53	450	477	0.00038	2
TSCO	-10.71	0.58**	-0.0039	-4.00	4.58	338	329	0.00061	2

Table A.8: Summary Result for the SMA Model

Ticker	BH	SMA	SR	Sell	Buy	#Buy	#Sell	Fast	Slow	Threshold
BLT	-18.23	11.78**	0.2330	-9.20	20.98	550	2958	3	6	0.00067
DGE	93.29	150.13**	0.7356	120.21	29.92	285	3133	2	5	0.00057
GSK	-28.61	30.22**	-0.0208	7.42	22.80	536	2884	2	5	0.00037
HSBA	46.05	49.00**	0.6769	49.93	-0.93	698	2817	2	6	0.00039
TSCO	-10.71	3.42**	0.1067	-3.76	7.18	420	3065	3	7	0.00061

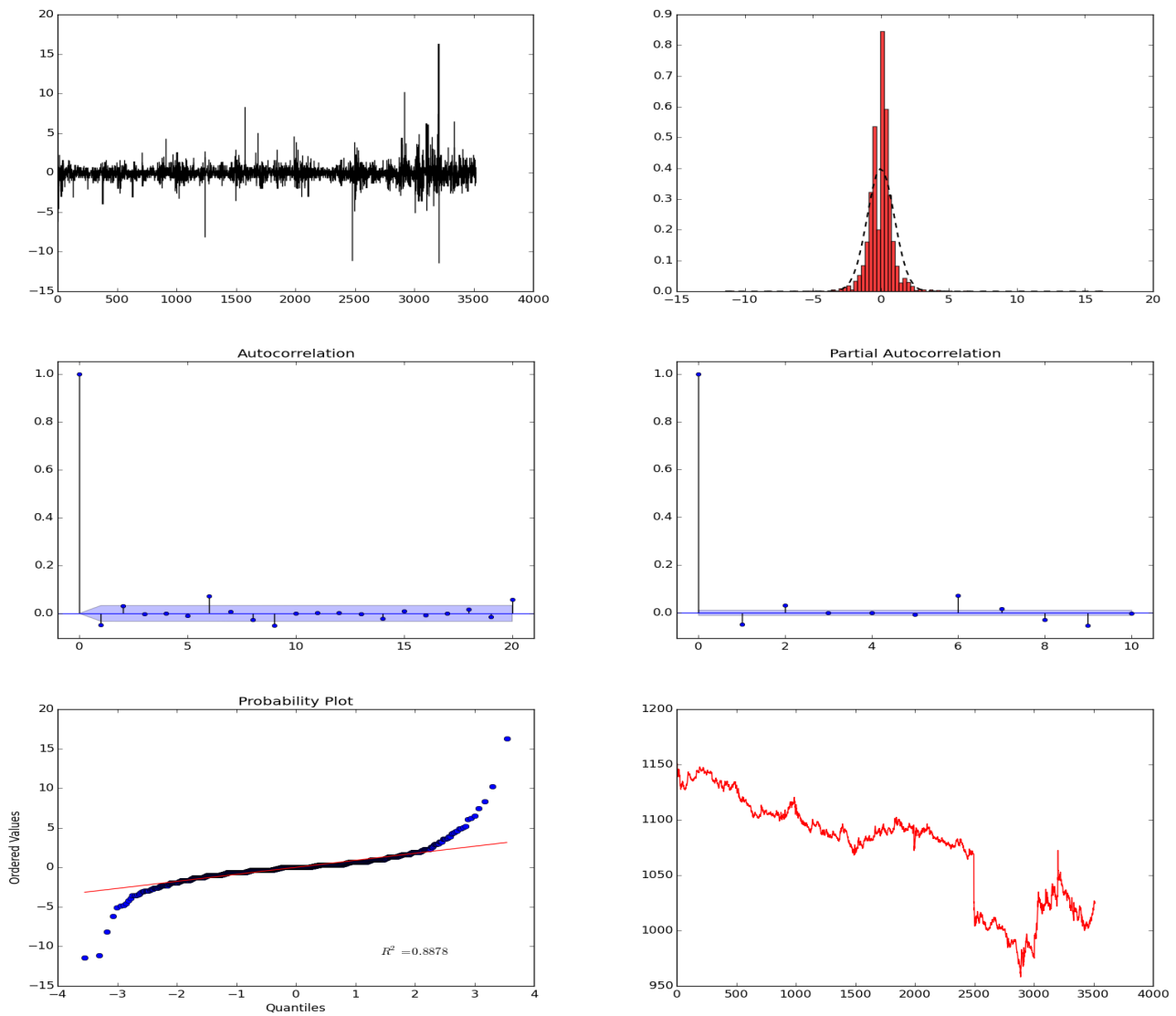
Table A.9: Summary of Performance by Date

Ticker	DATE	BH	EMA	KAL	MACD	RSI	SMPL1	SMPL2	SMA
BLT	24/08/2015	17.42	98.56	-9.71	27.75	9.42	45.13	28.27	57.84
	17/08/2015	30.3	41.46	34.67	24.03	25.37	-8.89	-0.73	55
	19/08/2015	40.08	9.44	39.16	52.18	20.6	-25.06	-35.56	11.02
	25/08/2015	-57.3	-17.62	60.48	99.2	-20.69	24.14	67.85	-67.3
	18/08/2015	-5.38	-25.48	-5.18	24.42	-22.69	-4	0.17	-11.74
	21/08/2015	8.23	-5.2	-37.21	-39.63	8.23	-14.04	-31.42	-1.76
	20/08/2015	-51.58	-56.4	-21.84	-20.06	-38.31	2.08	17.74	-31.28
Total		-18.23	44.76	60.37	167.89	-18.07	19.36	46.32	11.78
DGE	24/08/2015	28.03	105.71	75.53	112.83	76.46	63.93	64.92	89.25
	21/08/2015	34.4	24.54	-17.14	-16.96	36.77	-4.93	-6.9	43.76
	19/08/2015	15.27	6.15	22.83	10.43	8.05	-7.44	-3.63	12.99
	20/08/2015	12.25	8.67	-0.06	12	5.09	-0.14	2.23	10.23
	25/08/2015	1.89	7.07	13.22	31.66	4.17	-12.4	-4.8	5.93
	17/08/2015	7.28	14.5	-9.1	-0.92	3.57	-7.22	-5.57	-3.2
	18/08/2015	-5.83	-9.27	-24.31	-23.09	-6.43	5.92	5.01	-8.83
Total		93.29	157.37	60.97	125.95	127.68	37.72	51.26	150.13
GSK	24/08/2015	-8.82	71.1	29.29	66.28	38.5	38.15	47.28	68.14
	19/08/2015	23.13	18.49	4.3	17.84	15.83	1	-0.1	15.63
	21/08/2015	1.54	-0.02	-13.83	3.83	-2.27	0.31	-5.84	5.8
	20/08/2015	-4.42	-8.2	-14.44	-4.26	-2.38	-0.5	1.44	-8.52
	17/08/2015	-22.67	-24.15	19.75	13.37	-12.22	2.81	0.1	-22.01
	25/08/2015	-16.64	30.94	-48.3	-15.14	-2.21	0.31	-18.46	7.06
	18/08/2015	-0.73	-43.46	11.15	-10.92	1.96	-14.15	-13.58	-35.88
Total		-28.61	44.7	-12.08	71	37.21	27.93	10.84	30.22
HSBA	24/08/2015	37.95	53.17	25.21	38.81	30.69	23.58	-6.46	37.41
	20/08/2015	8.22	12.58	3.61	6.73	12.68	0.2	0.13	14.6
	25/08/2015	-12.67	-2.15	7.81	14.83	-1.07	13.89	13.95	-6.27
	21/08/2015	8.68	6.08	-3.29	4.95	10.35	-3.08	-2.62	7.04
	19/08/2015	9.41	6.6	8.1	5.16	1.81	-6.22	-5.3	4.92
	17/08/2015	0.59	-2.25	-1.27	-4.01	2.23	-2.57	0.18	-0.87
	18/08/2015	-6.13	-6.79	-3.2	-0.4	-4.71	-0.07	1.25	-7.83
Total		46.05	67.24	36.97	66.07	51.98	25.73	1.13	49
TSCO	24/08/2015	-4.53	3.65	6.69	4.82	3.78	3.83	-0.39	2.85
	20/08/2015	-4.16	3.18	5.36	7.02	0.34	-0.44	0.77	4.36
	19/08/2015	1.66	-0.84	2.1	7.11	0.11	0.08	0.58	0.4
	18/08/2015	0.94	-1.02	-7.6	2.59	1.91	1.51	-0.12	-0.04
	17/08/2015	0.2	0.91	-2.6	-0.02	-0.95	-0.82	-0.54	0.25
	21/08/2015	-0.48	-0.1	3.04	-2.47	-1.62	-1.57	-0.55	-0.86
	25/08/2015	-4.34	-1.68	2.1	-1.69	-2.53	0.18	0.83	-3.54
Total		-10.71	4.1	9.09	17.36	1.04	2.77	0.58	3.42
Grand Total		81.79	318.17	155.32	448.27	199.84	113.51	110.13	244.55

A.4 TEST SET STATISTICS AND GRAPHS

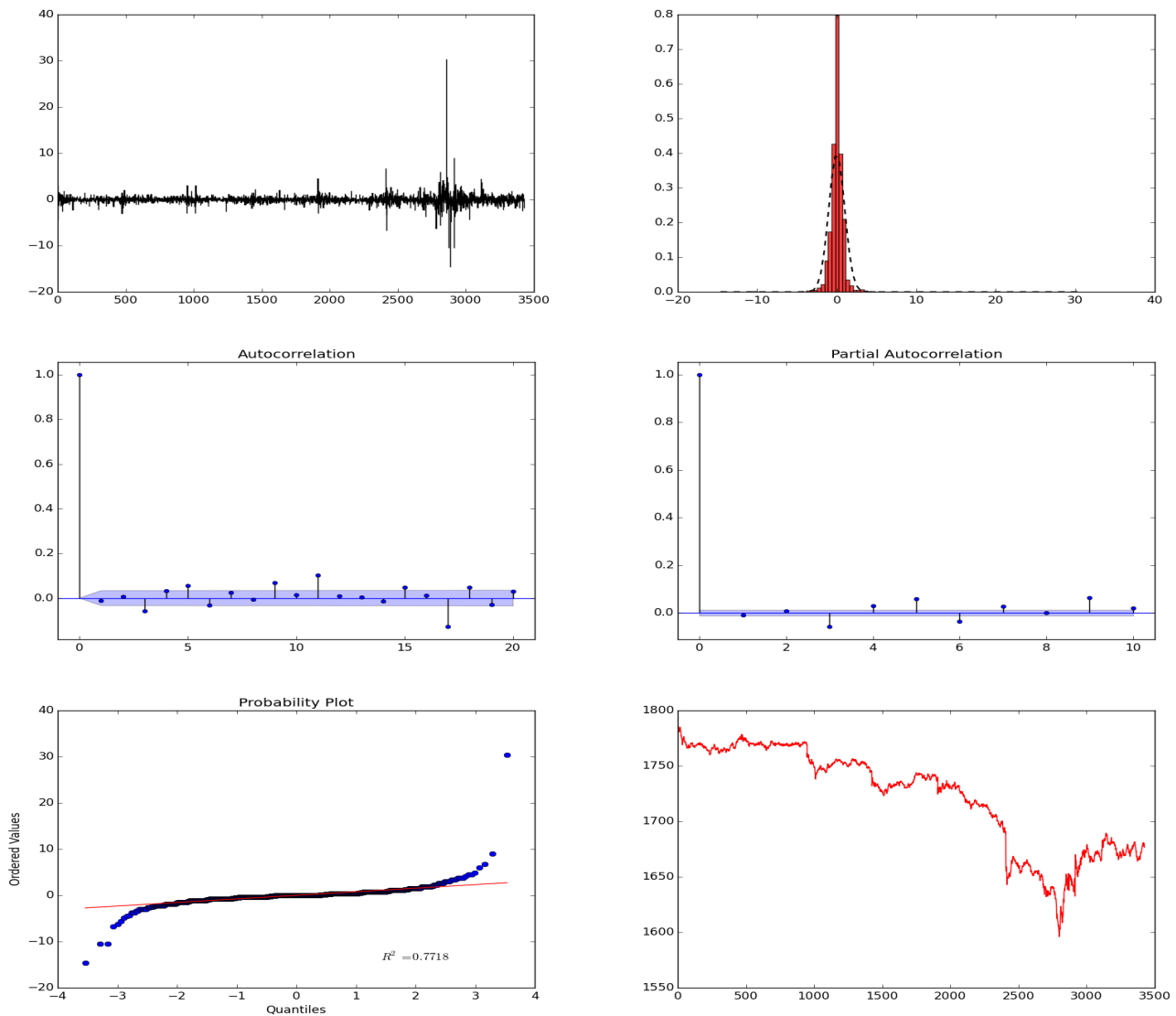
Tickers	Date	Time Ticks	Mean	Std	Skew	Kurtosis	JB Stat	DW Stat
BLT	17/08/2015	495	-0.0612	1.1537	-1.45	21	6573	2.12
	18/08/2015	492	0.0109	0.9950	0.41	9	718	1.93
	19/08/2015	504	-0.0795	1.1678	-2.26	29	14906	2.36
	20/08/2015	503	0.1025	1.3149	2.40	24	9539	1.91
	21/08/2015	502	-0.0164	1.3013	-4.31	62	74095	2.21
	24/08/2015	509	-0.0342	1.6370	1.97	22	7726	1.71
	25/08/2015	510	0.1124	2.5637	1.58	25	10873	2.18
DGE	17/08/2015	468	-0.0156	0.7429	-1.24	15	3072	2.16
	18/08/2015	482	0.0121	0.6265	-0.54	8	484	2.06
	19/08/2015	475	-0.0321	0.7113	-0.20	12	1540	2.10
	20/08/2015	484	-0.0253	0.6828	0.13	5	122	1.90
	21/08/2015	499	-0.0689	0.8103	0.66	11	1238	2.02
	24/08/2015	509	-0.0551	2.8163	4.90	93	174624	1.95
	25/08/2015	508	-0.0037	1.2730	0.05	6	165	2.16
GSK	17/08/2015	458	0.0495	0.6394	0.69	14	2545	2.01
	18/08/2015	488	0.0015	0.9472	-0.94	129	321495	2.77
	19/08/2015	477	-0.0485	0.5446	-1.48	12	1749	2.00
	20/08/2015	487	0.0091	0.5014	0.29	5	118	1.97
	21/08/2015	497	-0.0031	0.6408	0.17	6	166	2.00
	24/08/2015	510	0.0173	1.5303	2.03	19	5482	1.57
	25/08/2015	510	0.0326	1.0580	-0.01	7	295	2.10
HSBA	17/08/2015	494	-0.0012	0.2187	0.02	7	264	2.12
	18/08/2015	499	0.0123	0.2014	0.30	6	211	1.93
	19/08/2015	500	-0.0188	0.2524	-0.44	8	570	2.25
	20/08/2015	506	-0.0162	0.2489	-0.07	9	839	1.91
	21/08/2015	503	-0.0173	0.2722	0.09	4	43	2.15
	24/08/2015	510	-0.0744	0.7496	-2.90	43	34569	1.91
	25/08/2015	510	0.0248	0.4642	0.63	6	239	1.83
TSCO	17/08/2015	477	-0.0004	0.0932	0.23	6	132	2.09
	18/08/2015	494	-0.0019	0.1319	2.91	39	27179	1.93
	19/08/2015	504	-0.0033	0.1080	-0.69	8	534	1.89
	20/08/2015	501	0.0083	0.1688	3.64	36	23809	1.93
	21/08/2015	500	0.0010	0.1128	-0.01	6	251	2.01
	24/08/2015	507	0.0089	0.2263	-0.07	7	291	1.89
	25/08/2015	509	0.0085	0.1606	0.84	7	442	1.98

Table A.10: Summary Time Ticks

Figure A.7: BLT (BHP Billiton) Test Set

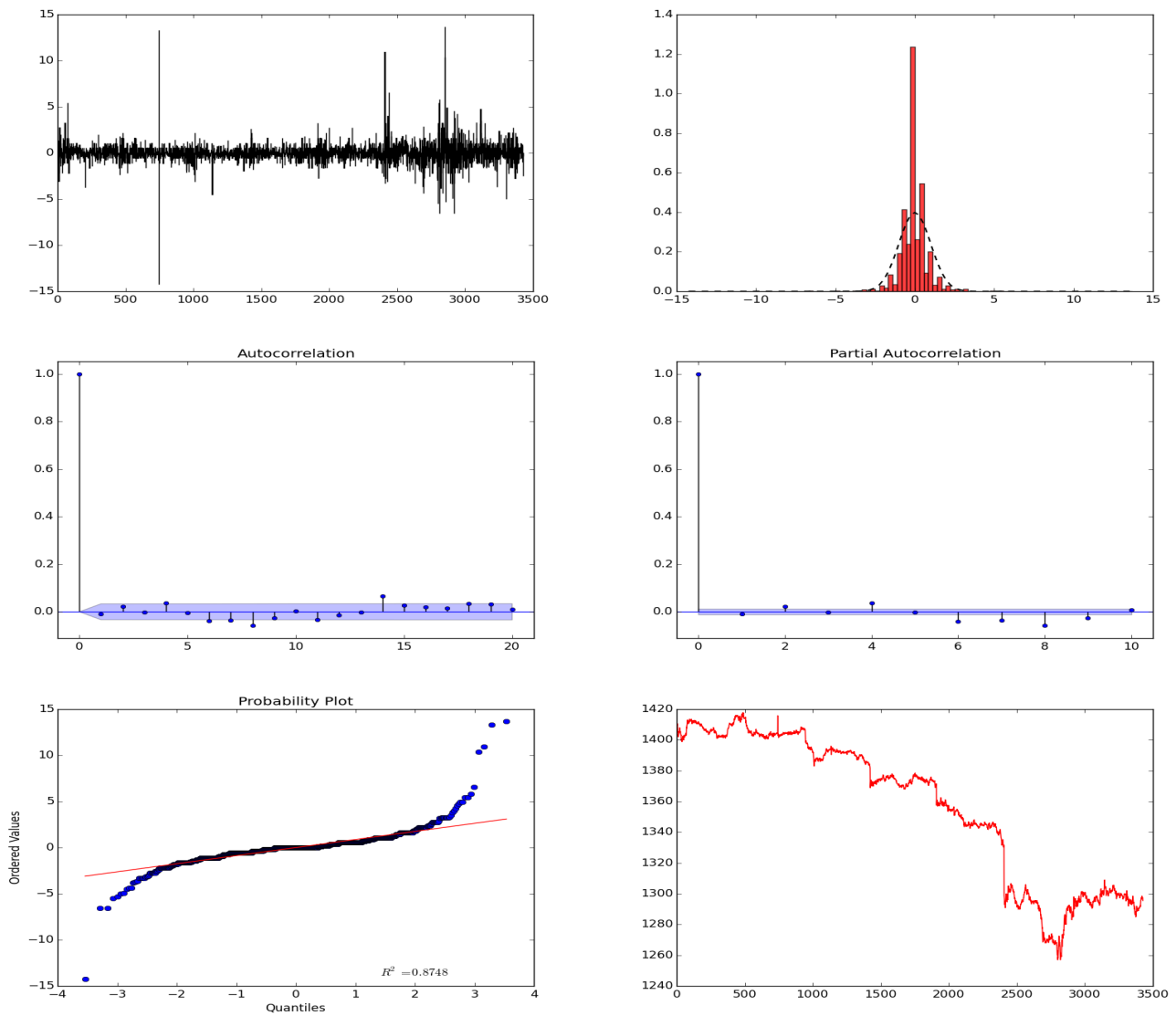
Figures starting from the top left hand corner ...

- Return plot (Open - Close)
- Normalized histogram of the returns
- Autocorrelation function of the returns
- Partial autocorrelation function of the returns
- Quantile-Quantile(QQ) Plot showing the horizontal red line as the normal distribution and the blue dots are the returns plotted against it.
- Simple plot of the closing prices of the Stock

Figure A.8: DGE (Diageo) Test Set

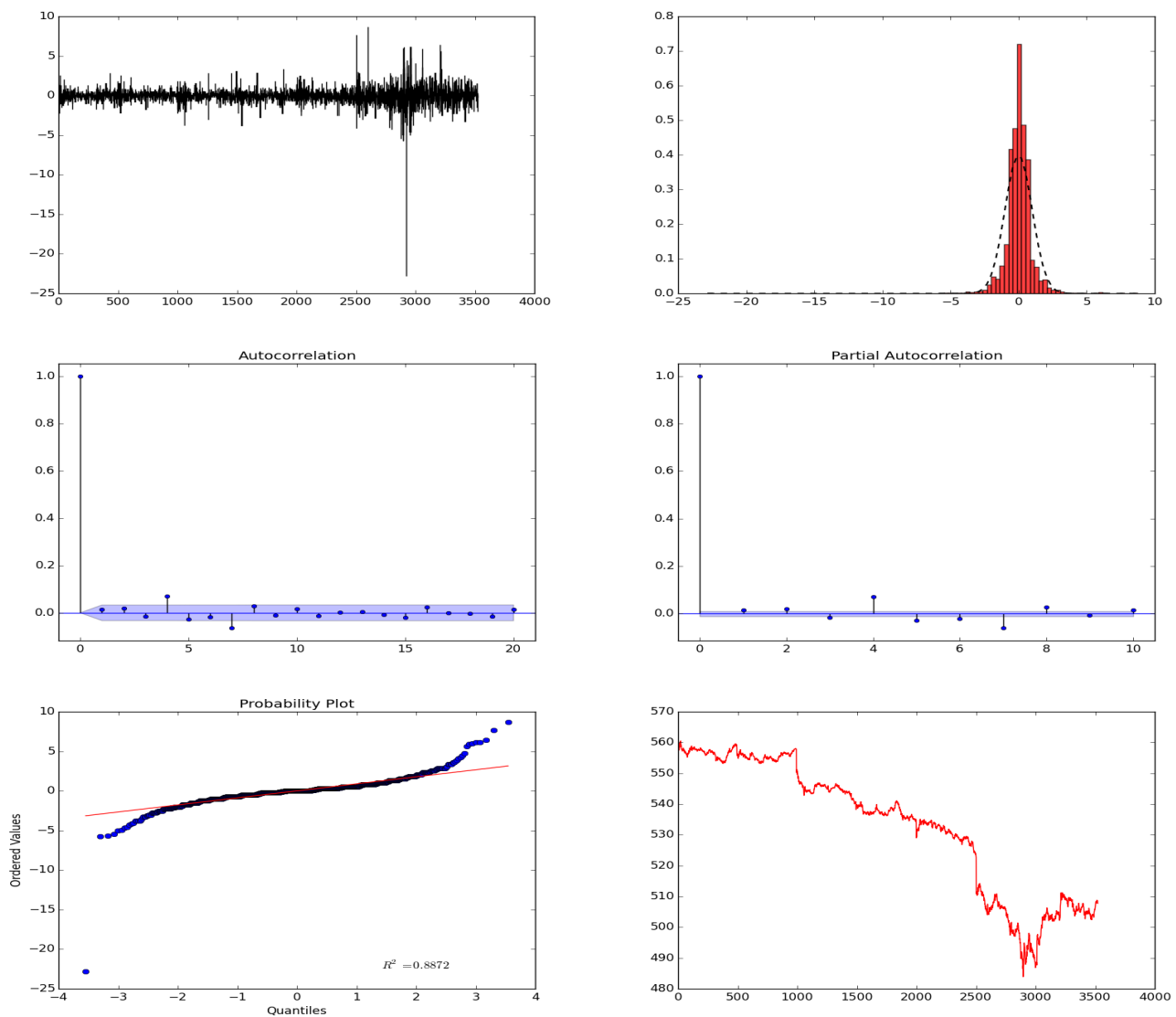
Figures starting from the top left hand corner ...

- Return plot (Open - Close)
- Normalized histogram of the returns
- Autocorrelation function of the returns
- Partial autocorrelation function of the returns
- Quantile-Quantile(QQ) Plot showing the horizontal red line as the normal distribution and the blue dots are the returns plotted against it.
- Simple plot of the closing prices of the Stock

Figure A.9: GSK(GlaxoSmithKline) Test Set

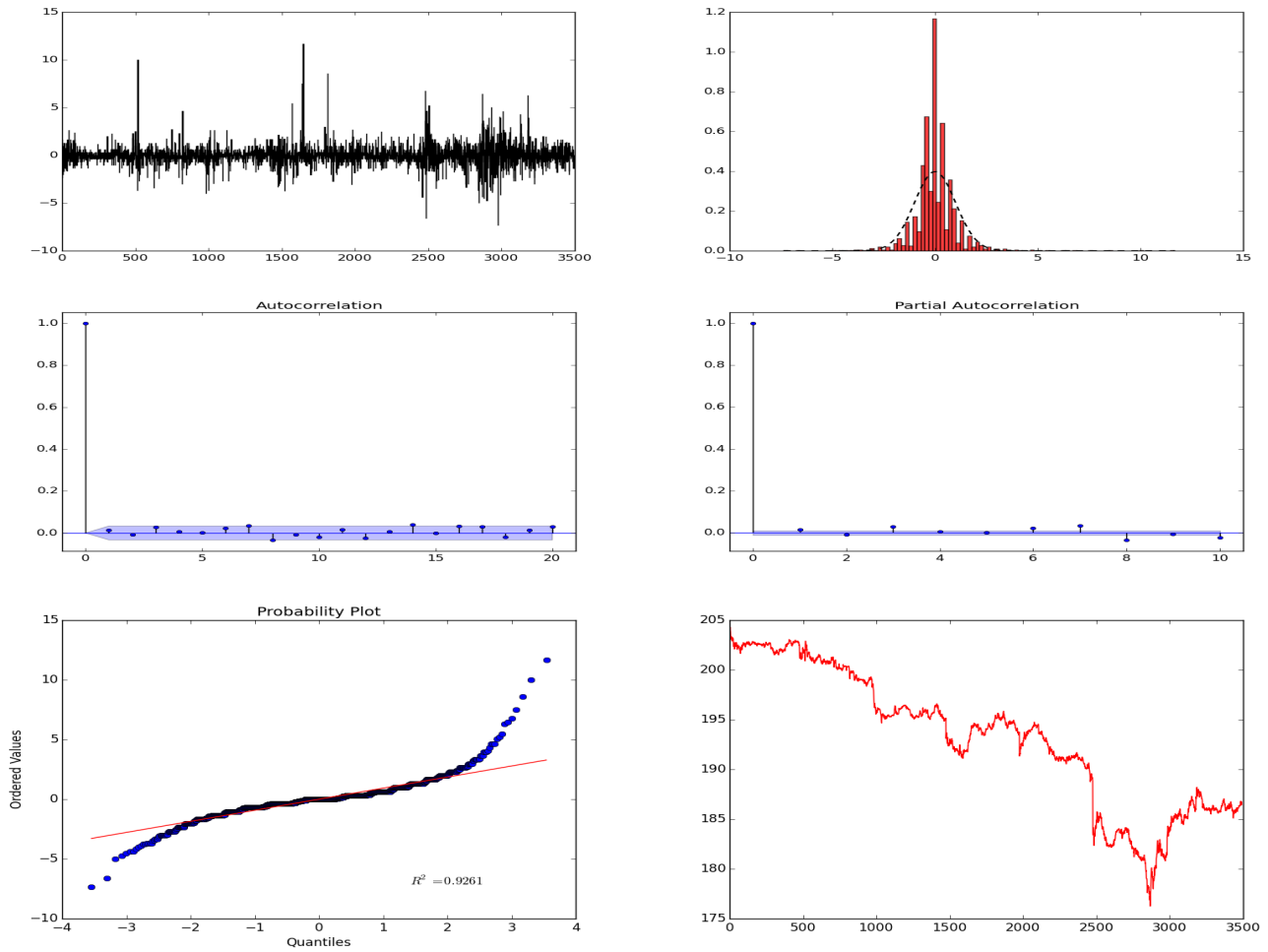
Figures starting from the top left hand corner ...

- Return plot (Open - Close)
- Normalized histogram of the returns
- Autocorrelation function of the returns
- Partial autocorrelation function of the returns
- Quantile-Quantile(QQ) Plot showing the horizontal red line as the normal distribution and the blue dots are the returns plotted against it.
- Simple plot of the closing prices of the Stock

Figure A.10: HSBA (HSBC Holdings) Test Set

Figures starting from the top left hand corner ...

- Return plot (Open - Close)
- Normalized histogram of the returns
- Autocorrelation function of the returns
- Partial autocorrelation function of the returns
- Quantile-Quantile(QQ) Plot showing the horizontal red line as the normal distribution and the blue dots are the returns plotted against it.
- Simple plot of the closing prices of the Stock

Figure A.11: TSCO (Tesco) Test Set

Figures starting from the top left hand corner ...

- Return plot (Open - Close)
- Normalized histogram of the returns
- Autocorrelation function of the returns
- Partial autocorrelation function of the returns
- Quantile-Quantile(QQ) Plot showing the horizontal red line as the normal distribution and the blue dots are the returns plotted against it.
- Simple plot of the closing prices of the Stock

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